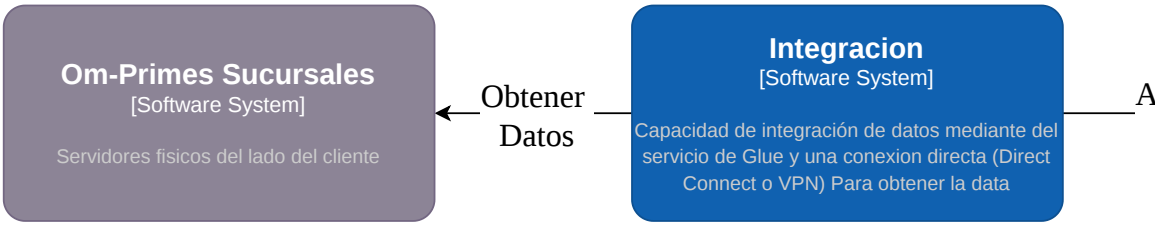
Reto: Estrategia Integral para la Adopción de la Nube y Transformación Digital en el Sector Bancario

Descripción general del contexto

Banca Global S.A.S., empresa financiera líder en América Latina, decidió implementar soluciones en la nube para optimizar sus operaciones tecnológicas y acelerar su transformación digital. Hoy en día, la banca global gestiona la infraestructura local utilizando servidores y contenedores físicos que admiten aplicaciones críticas como gestión de cuentas, pagos, análisis de datos y detección de fraudes.

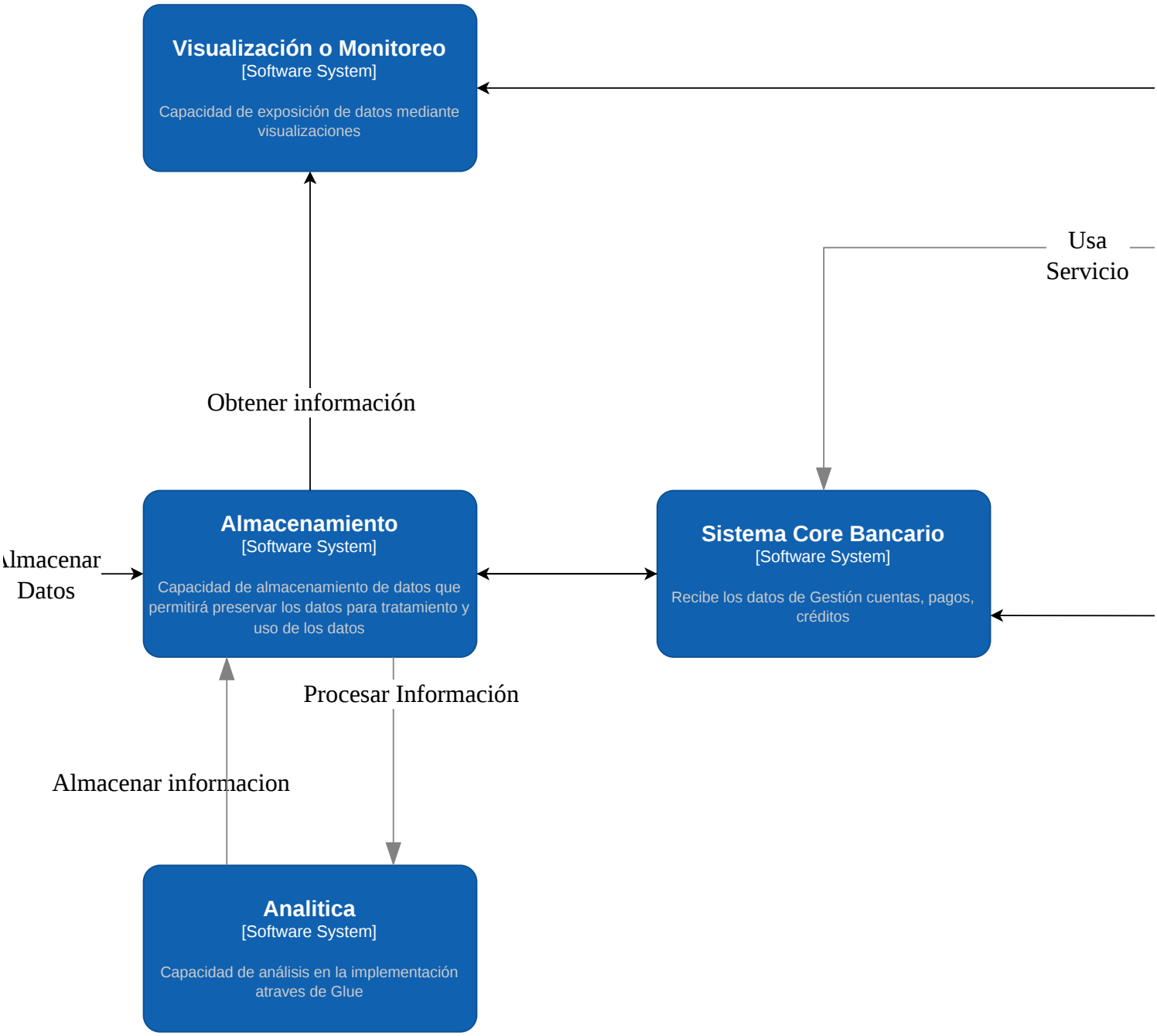
Objetivo del Reto

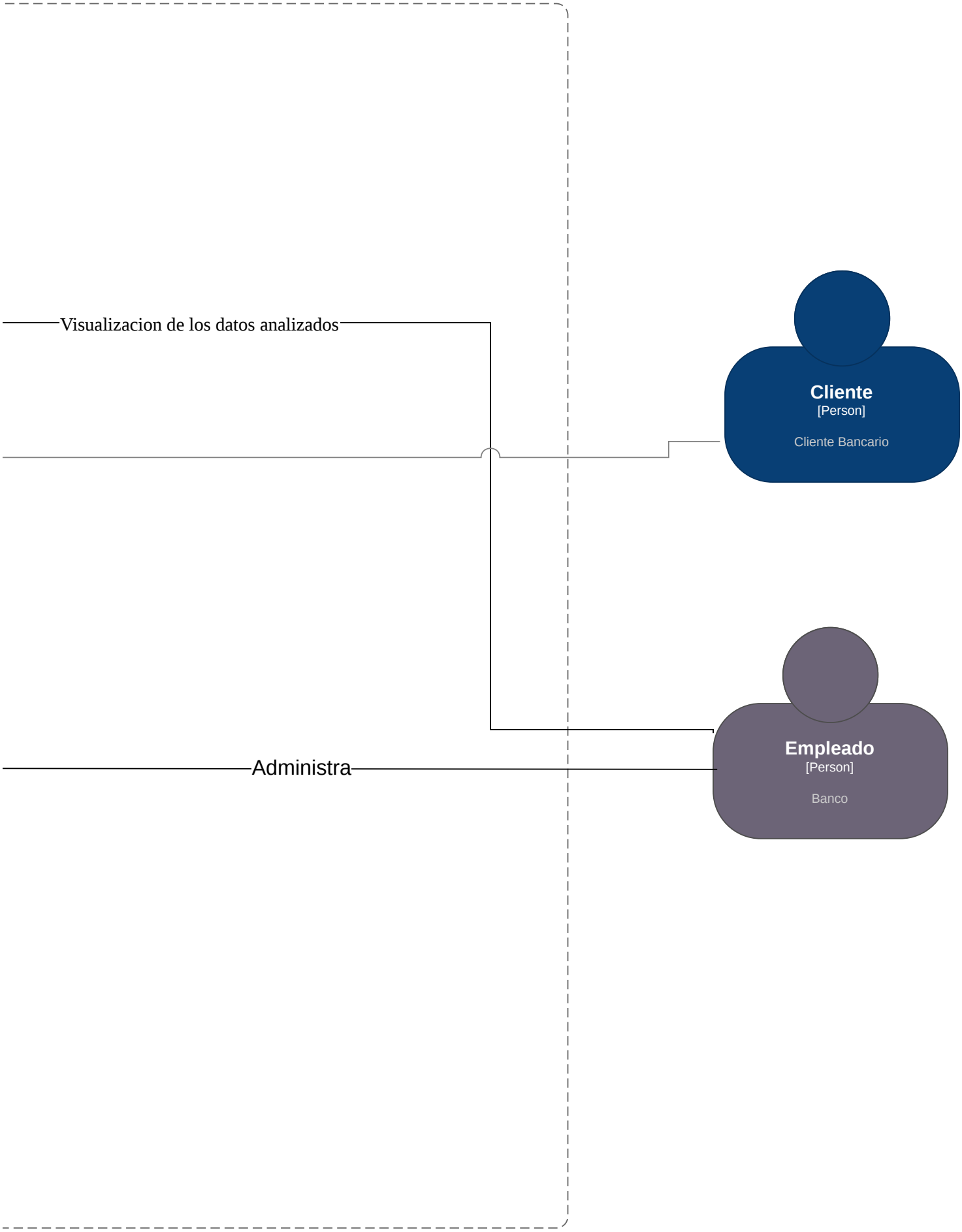
Diseñar y sustentar una estrategia integral para la adopción de la nube que contemple las siguientes áreas clave: Gobernanza, Seguridad, Networking, DevOps y la Implementación de Aplicaciones Nativas en la Nube, así como el procesamiento de datos en tiempo real. Esta estrategia debe alinearse con los principios de los Well-Architected Frameworks de las principales plataformas en la nube, que incluyen aspectos como Excelencia Operacional, Seguridad, Confiabilidad, Eficiencia


Om-Primes Sucursales
[Software System]
Servidores fisicos del lado del cliente
Obtener Datos
Integracion
[Software System]
Capacidad de integración de datos mediante del servicio de Glue y una conexion directa (Direct Connect o VPN) Para obtener la data
A

[]

Banca Global S.A.S




Visualizacion de los datos analizados
Cliente
[Person]
Cliente Bancario
Empleado
[Person]
Banco
Administra

Banca Global S.A.S

Wep Application
[[Container: Amazon S3 y CloudFr

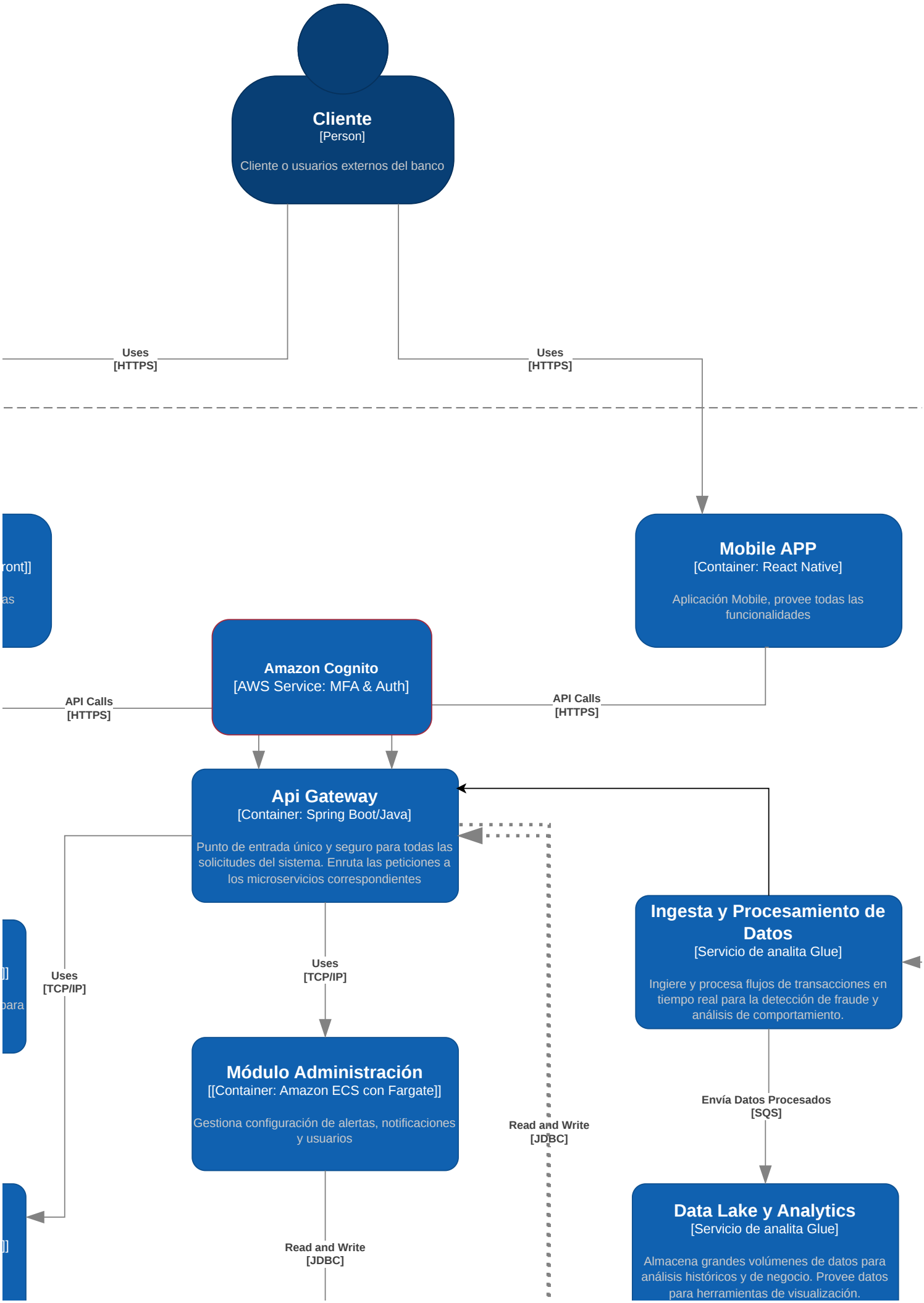
Sitio Web estático, provee todas la
funcionalidades

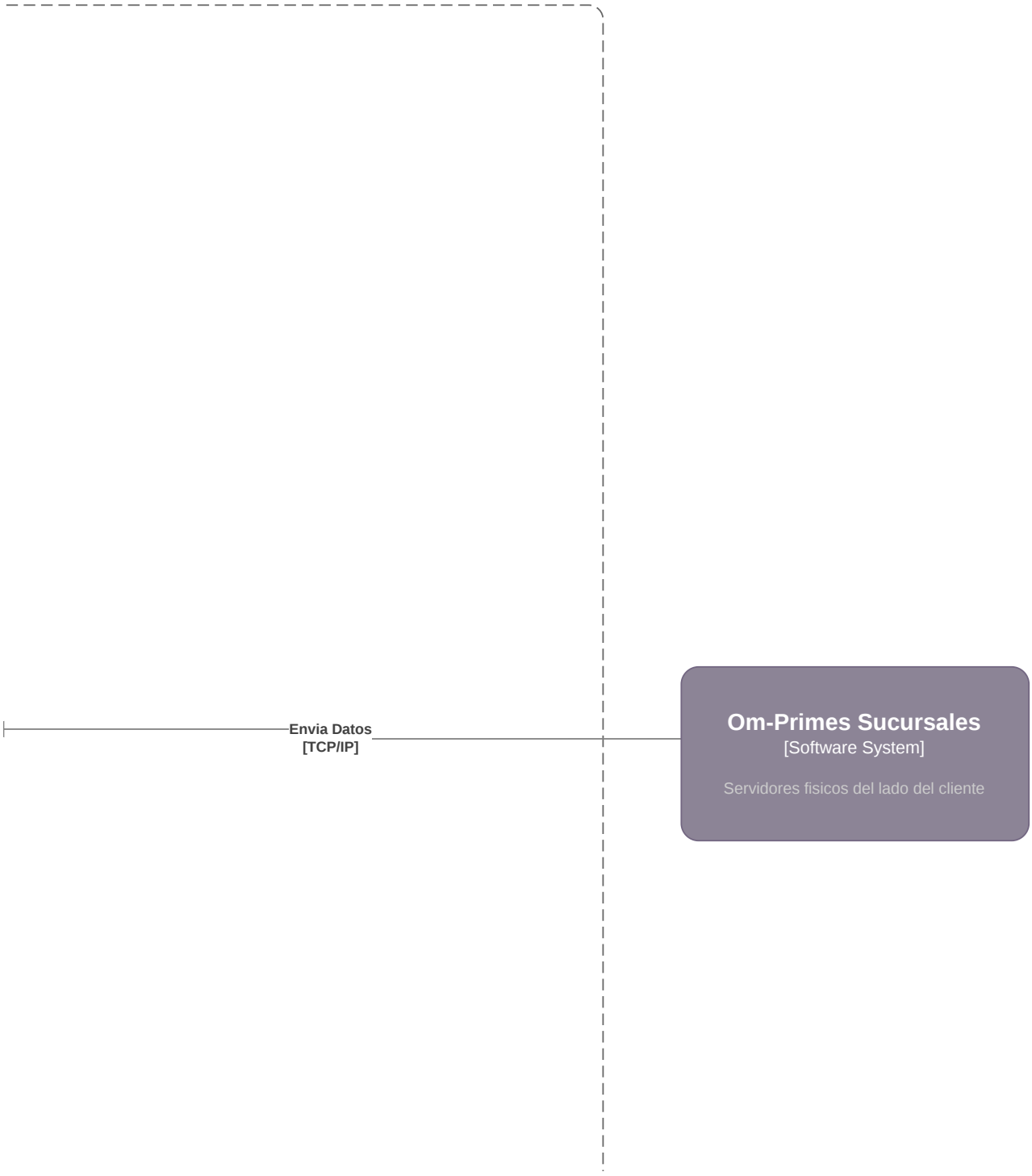
Módulo Créditos
[[Container: Amazon ECS con Fargate]

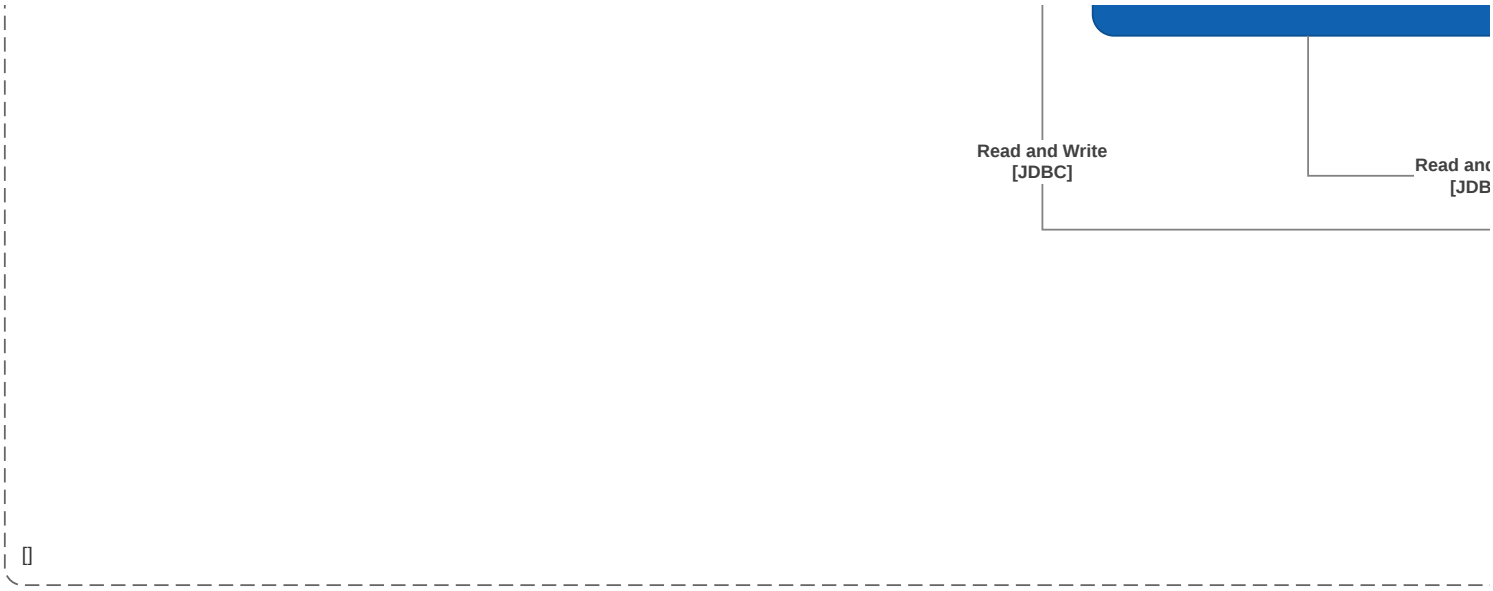
Gestiona toda la informacion de los clientes p
validacion de creditos

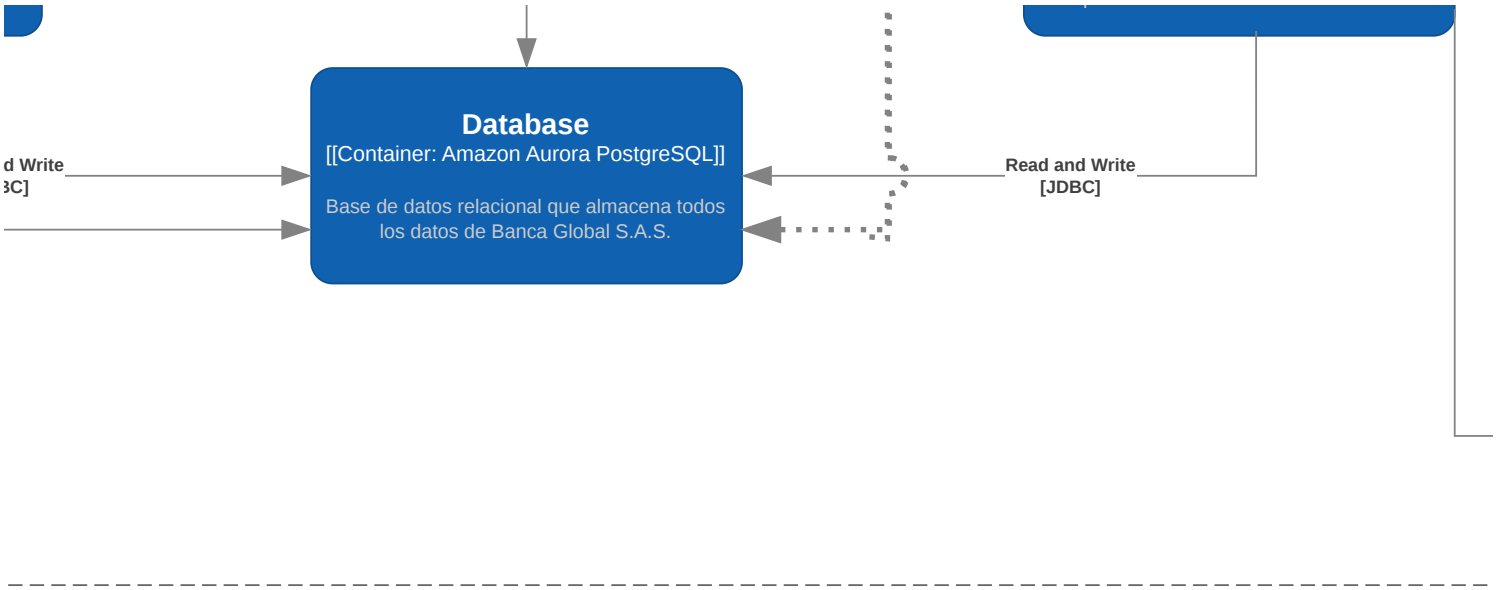
Módulo Recaudo
[[Container: Amazon ECS con Fargate]

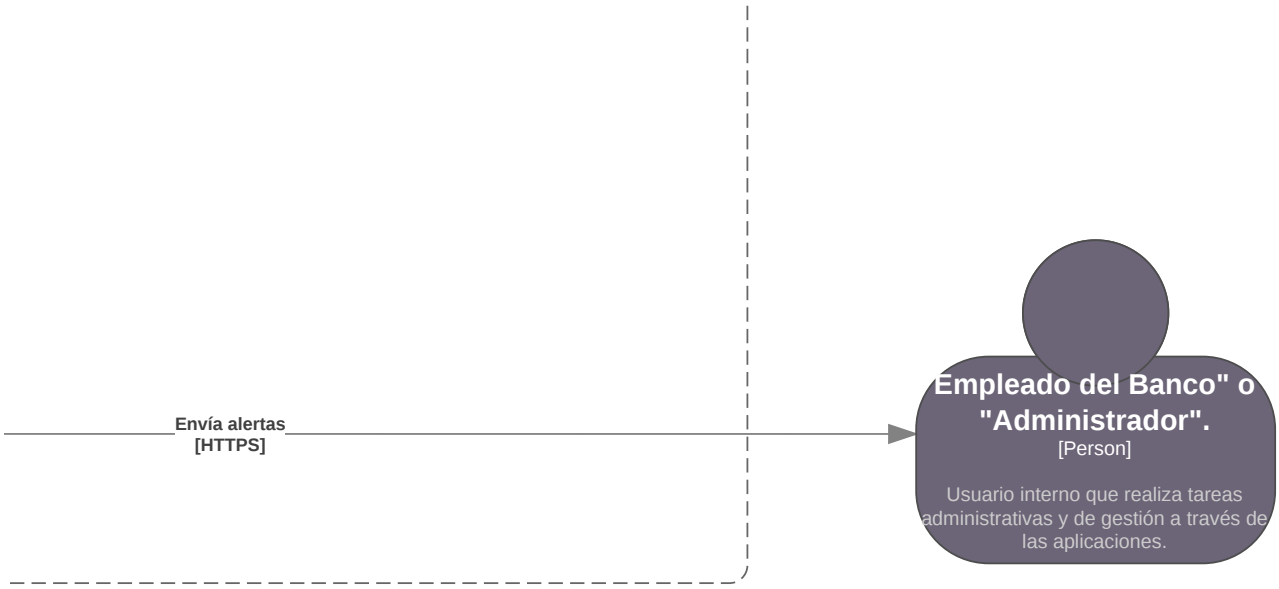
Gestiona facturación y procesa pagos

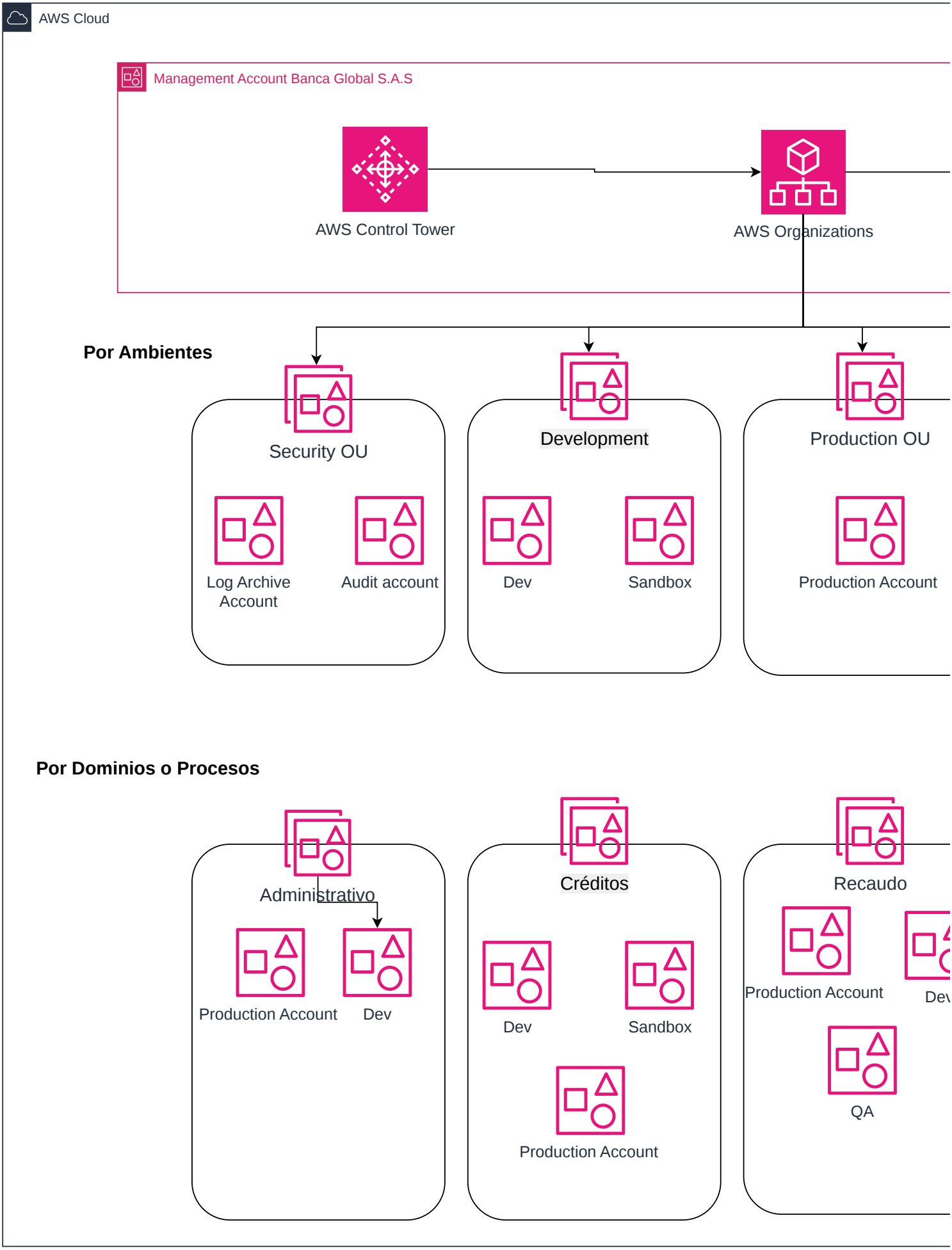

Cliente
[Person]
Cliente o usuarios externos del banco
Uses
[HTTPS]
Uses
[HTTPS]
Mobile APP
[Container: React Native]
Aplicación Mobile, provee todas las funcionalidades
Amazon Cognito
[AWS Service: MFA & Auth]
API Calls
[HTTPS]
API Calls
[HTTPS]
Api Gateway
[Container: Spring Boot/Java]
Punto de entrada único y seguro para todas las solicitudes del sistema. Enruta las peticiones a los microservicios correspondientes
Uses
[TCP/IP]
Uses
[TCP/IP]
Ingesta y Procesamiento de Datos
[Servicio de analita Glue]
Ingiere y procesa flujos de transacciones en tiempo real para la detección de fraude y análisis de comportamiento.
Módulo Administración
[[Container: Amazon ECS con Fargate]]
Gestiona configuración de alertas, notificaciones y usuarios
Read and Write
[JDBC]
Envía Datos Procesados
[SQS]
Read and Write
[JDBC]
Data Lake y Analytics
[Servicio de analita Glue]
Almacena grandes volúmenes de datos para análisis históricos y de negocio. Provee datos para herramientas de visualización.

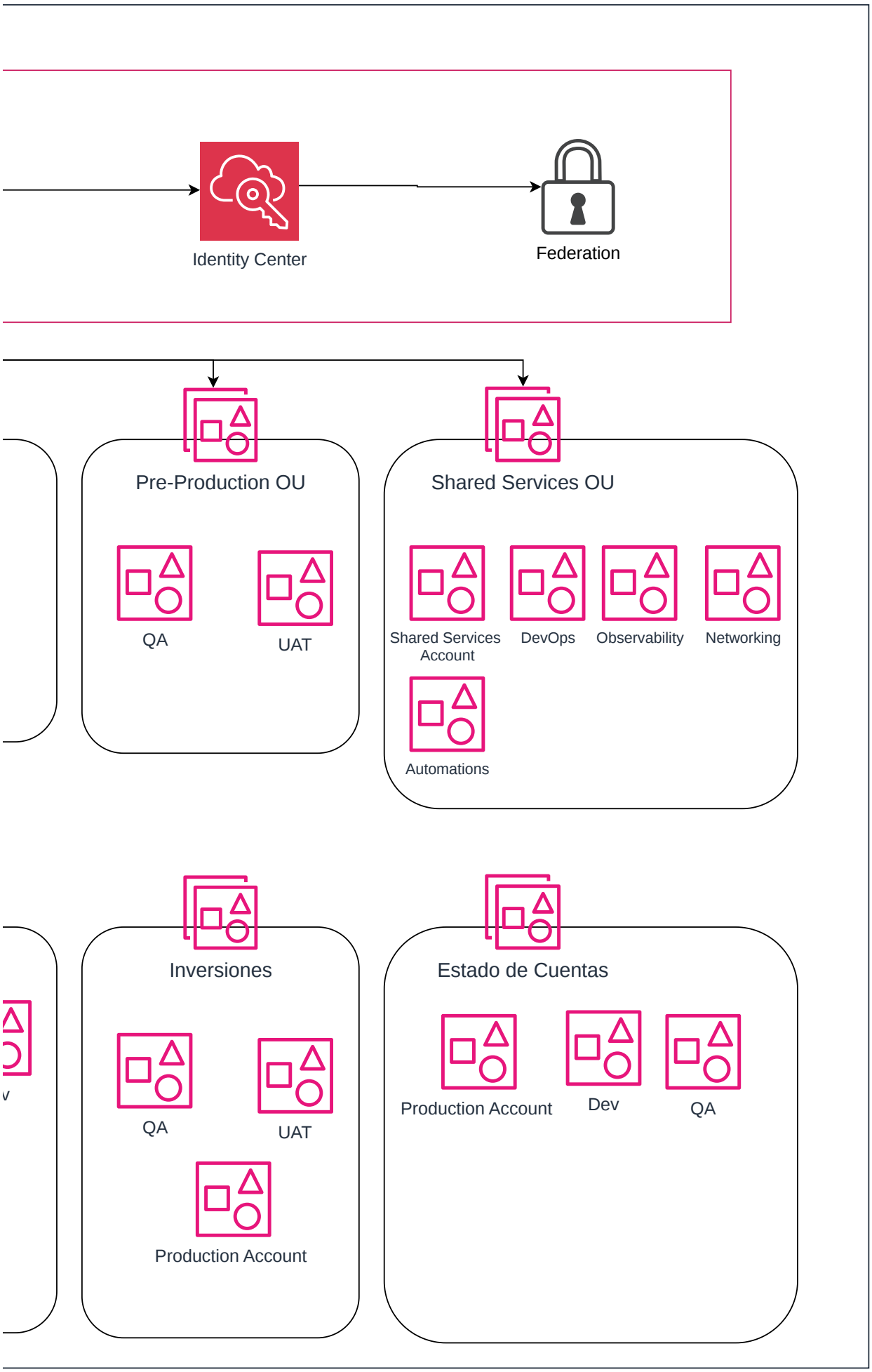

Envia Datos
[TCP/IP]
Om-Primes Sucursales
[Software System]
Servidores fisicos del lado del cliente


Read and Write
[JDBC]
Read and
[JDB


Database
[[Container: Amazon Aurora PostgreSQL]]
Base de datos relacional que almacena todos los datos de Banca Global S.A.S.
d Write
3C]
Read and Write
[JDBC]


Envía alertas
[HTTPS]
Empleado del Banco" o "Administrador".
[Person]
Usuario interno que realiza tareas administrativas y de gestión a través de las aplicaciones.


AWS Cloud
Management Account Banca Global S.A.S
AWS Control Tower
AWS Organizations
Por Ambientes
Security OU
Log Archive Account
Audit account
Development
Dev
Sandbox
Production OU
Production Account
Por Dominios o Procesos
Administrativo
Production Account
Dev
Créditos
Dev
Sandbox
Production Account
Recaudo
Production Account
Dev
QA


Identity Center
Federation
Pre-Production OU
QA
UAT
Shared Services OU
Shared Services Account
DevOps
Observability
Networking
Automations
Inversiones
QA
UAT
Production Account
Estado de Cuentas
Production Account
Dev
QA

- **The Security OU:** la cuenta de archivo de registro y la cuenta de auditoría. central para todos los registros de CloudTrail y AWS Config en la Landing
- **The Sandbox OU** está configurada para alojar cuentas de prueba (Cuentas carga de trabajo de producción.
- **Production OU –** esta unidad organizativa es responsable de alojar todas s
- **Pre-Production OU** – esta unidad organizativa se puede utilizar como ento
- **Shared Services OU** – las cuentas de los servicios de host de la unidad org otras cuentas.

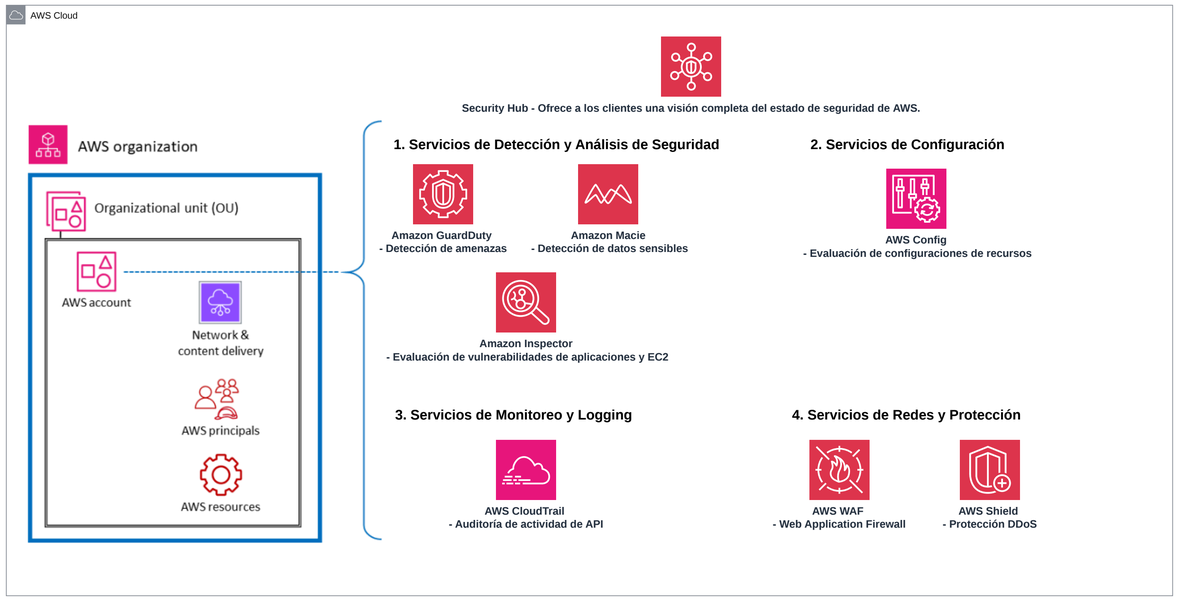
La cuenta de archivo de registros actúa como un repositorio
Zone, que se guardan de forma segura en un S3 Bucket.

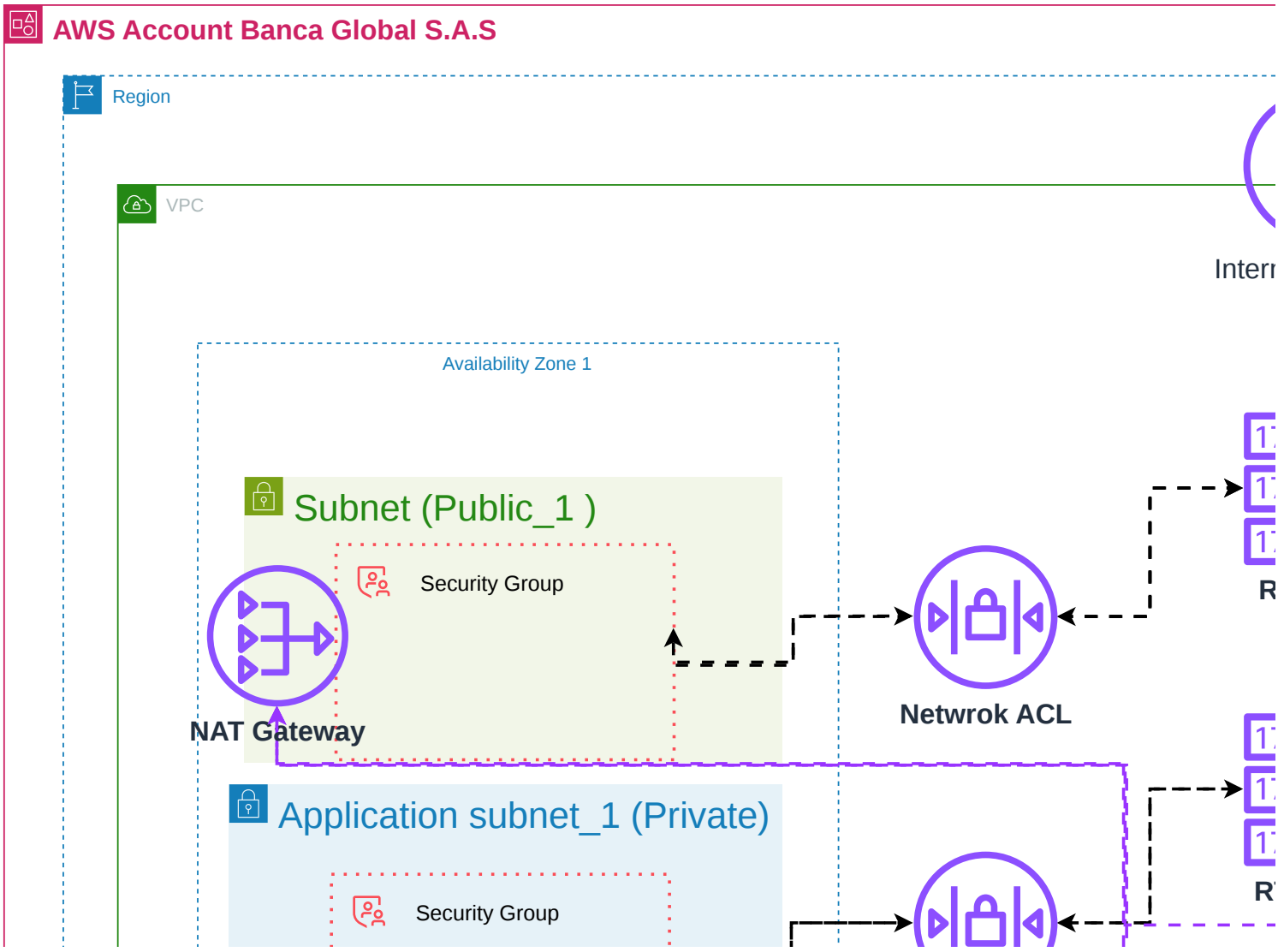
s Sandbox) que están segregadas de forma segura de cualquier

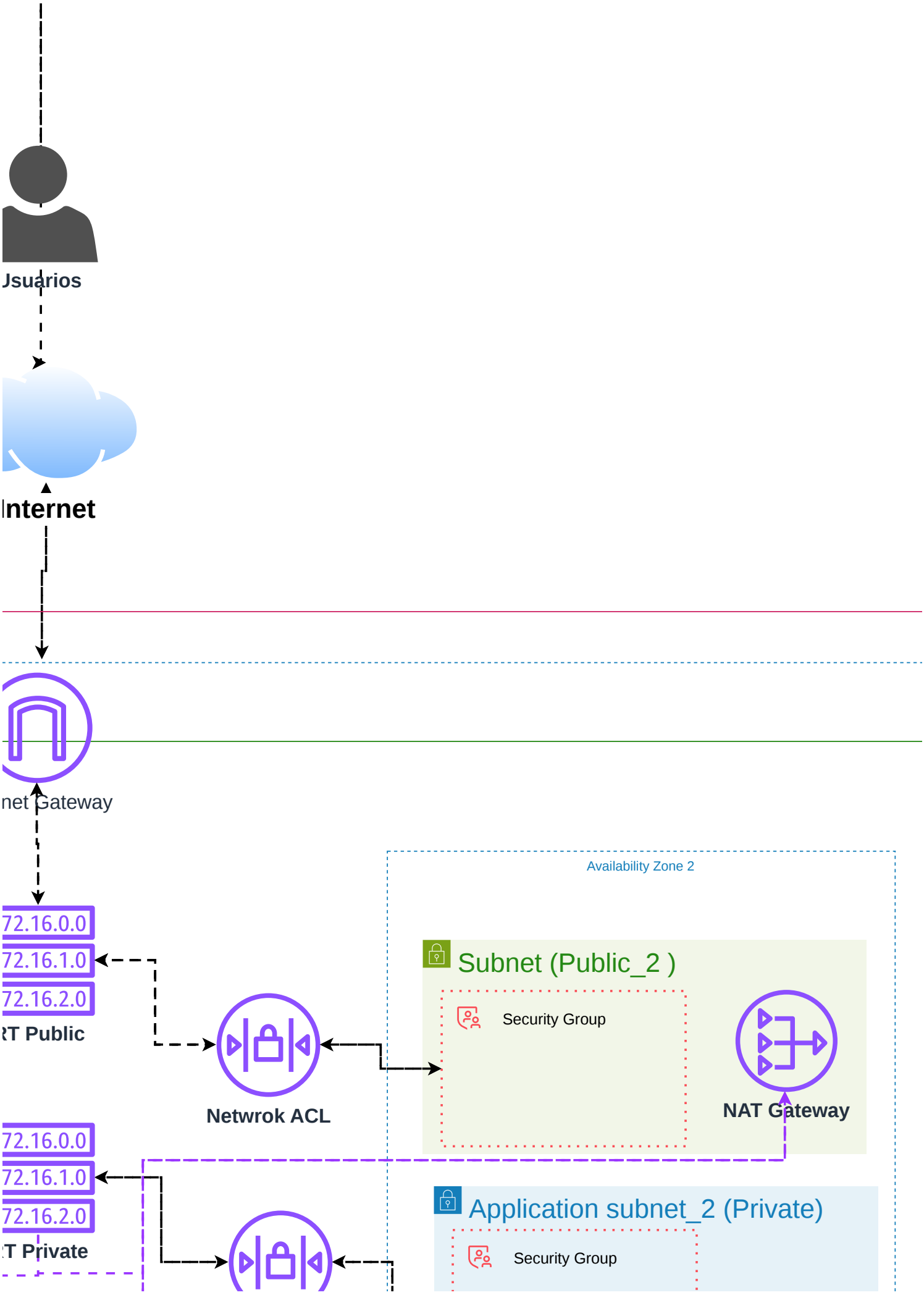
sus cuentas de producción y cargas de trabajo.

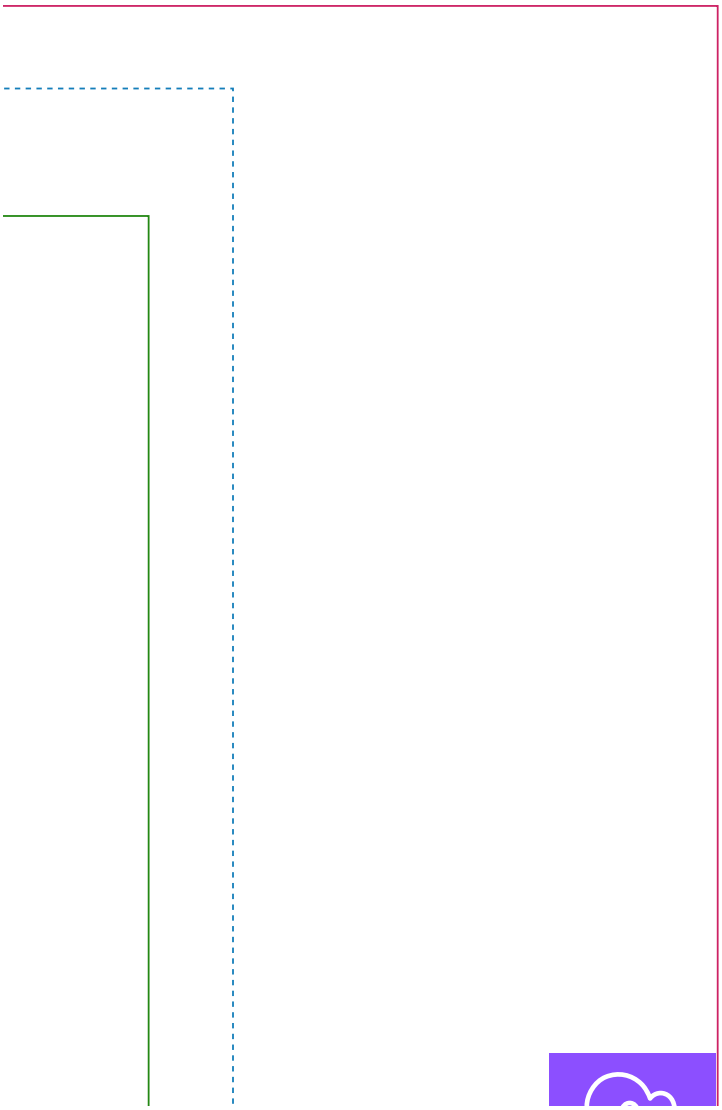
rno de preproducción para pruebas y desarrollo adicionales.

anizativa de servicios compartidos son compartidas por muchas

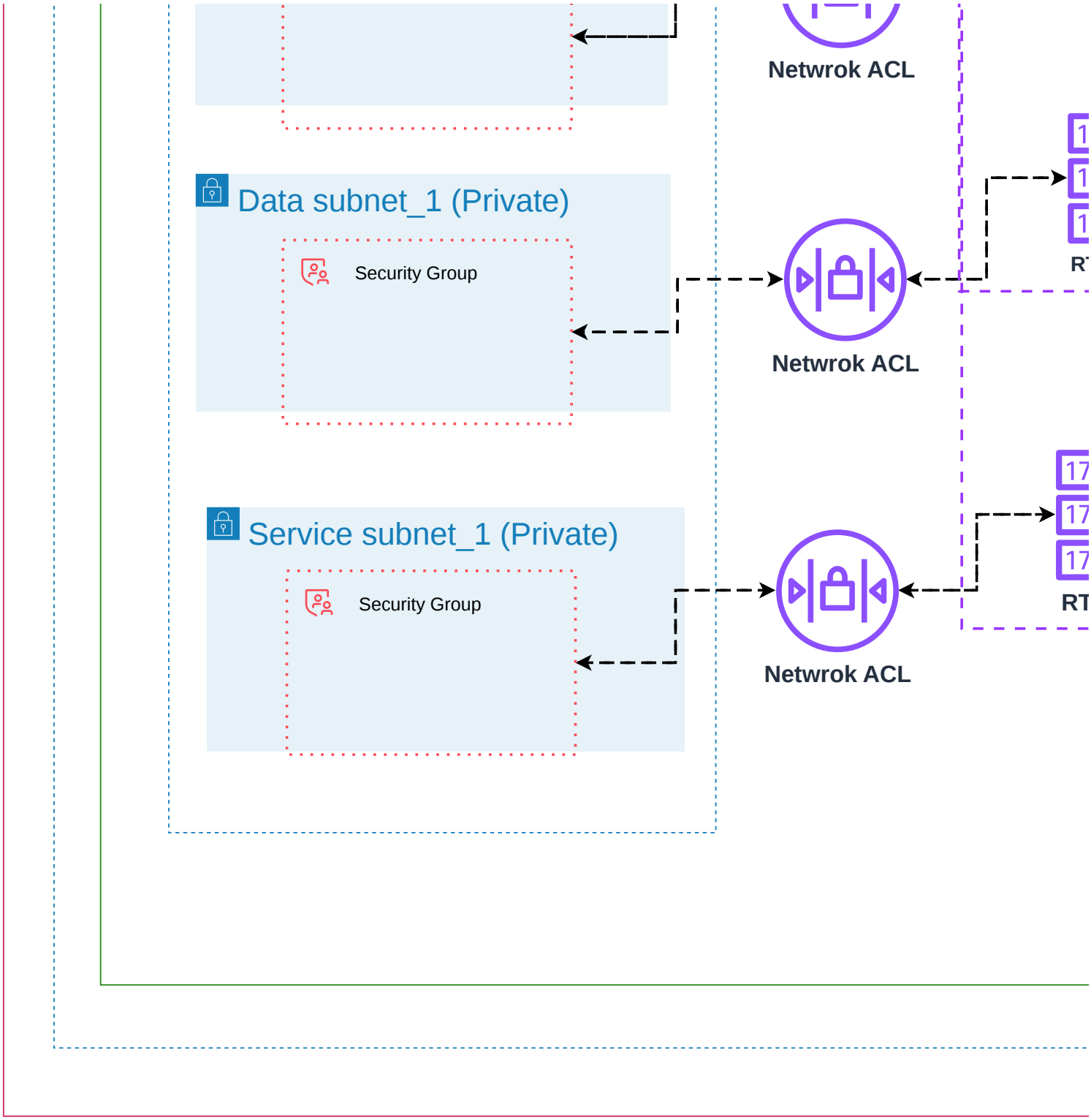

AWS Cloud
Security Hub - Ofrece a los clientes una visión completa del estado de seguridad de AWS.
AWS organization
Organizational unit (OU)
AWS account
Network & content delivery
AWS principals
AWS resources
1. Servicios de Detección y Análisis de Seguridad
Amazon GuardDuty
- Detección de amenazas
Amazon Macie
- Detección de datos sensibles
Amazon Inspector
- Evaluación de vulnerabilidades de aplicaciones y EC2
2. Servicios de Configuración
AWS Config
- Evaluación de configuraciones de recursos
3. Servicios de Monitoreo y Logging
AWS CloudTrail
- Auditoría de actividad de API
4. Servicios de Redes y Protección
AWS WAF
- Web Application Firewall
AWS Shield
- Protección DDoS

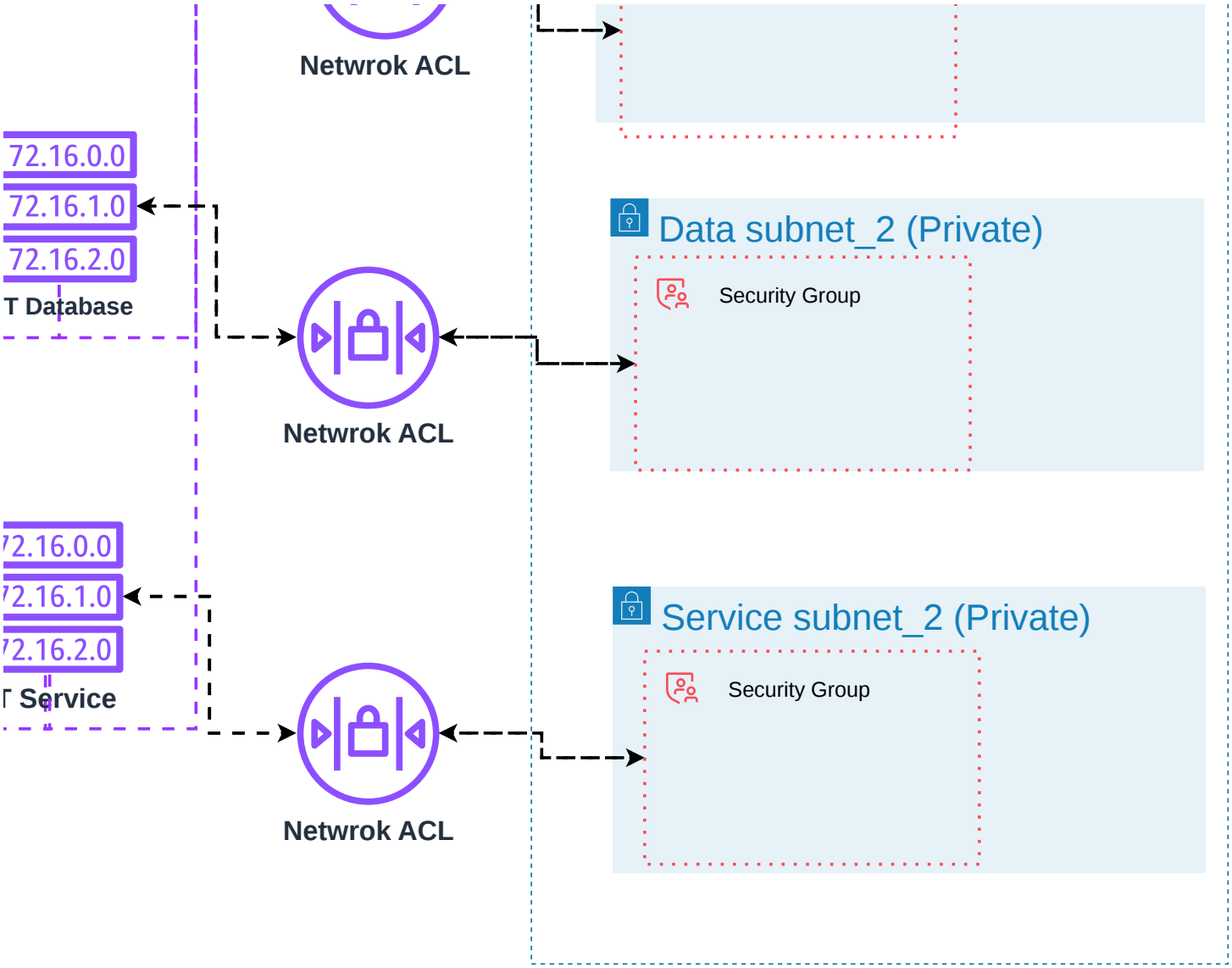

AWS Account Banca Global S.A.S
Region
VPC
Availability Zone 1
Subnet (Public_1)
Security Group
NAT Gateway
Netwrok ACL
Application subnet_1 (Private)
Security Group

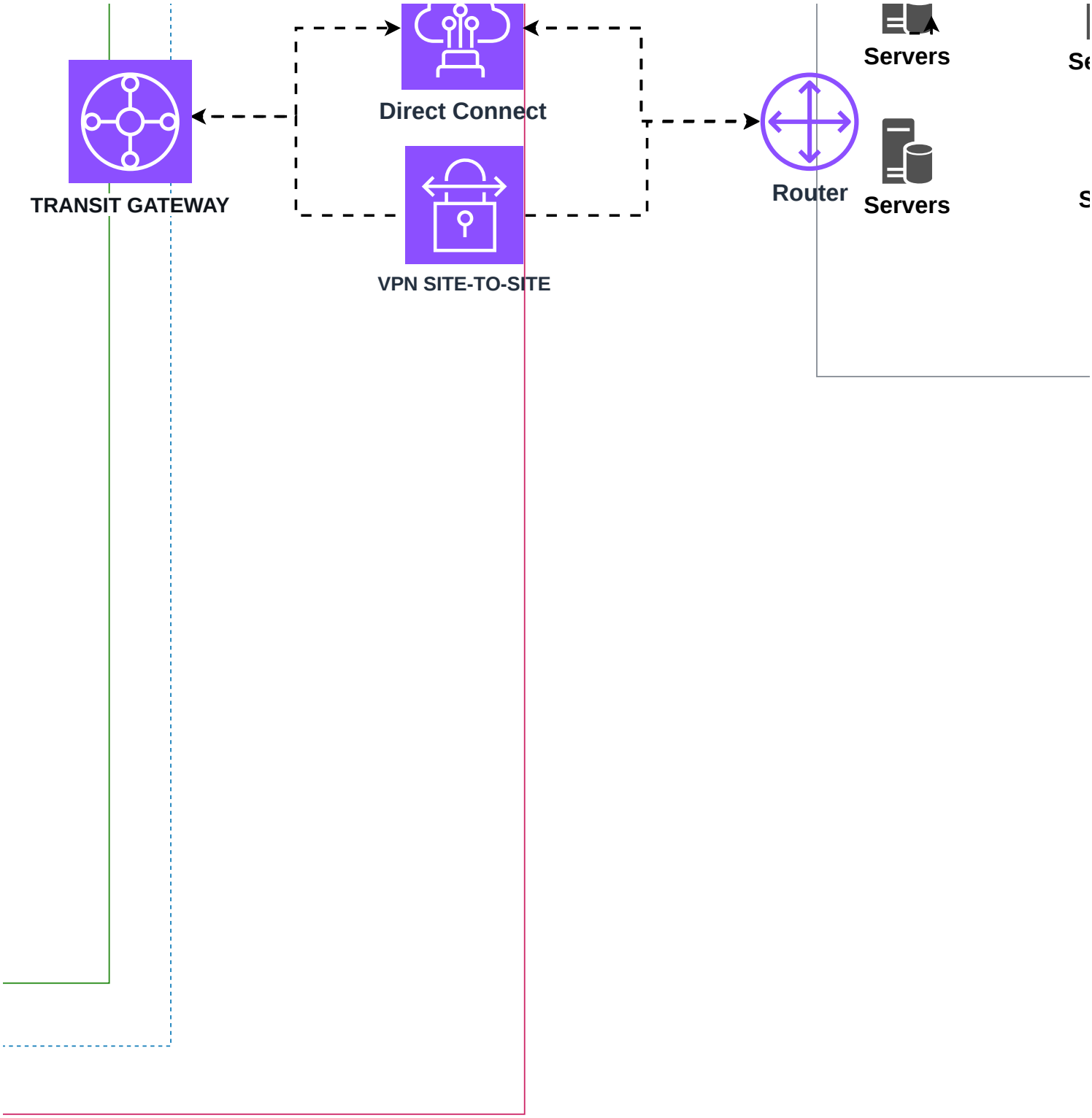

Jsuarios
Internet
net Gateway
Availability Zone 2
72.16.0.0
72.16.1.0
72.16.2.0
RT Public
Subnet (Public_2)
Security Group
Netwrok ACL
NAT Gateway
72.16.0.0
72.16.1.0
72.16.2.0
T Private
Application subnet_2 (Private)
Security Group



On-Premise Banca G


Netwrok ACL
Data subnet_1 (Private)
Security Group
Netwrok ACL
Service subnet_1 (Private)
Security Group
Netwrok ACL

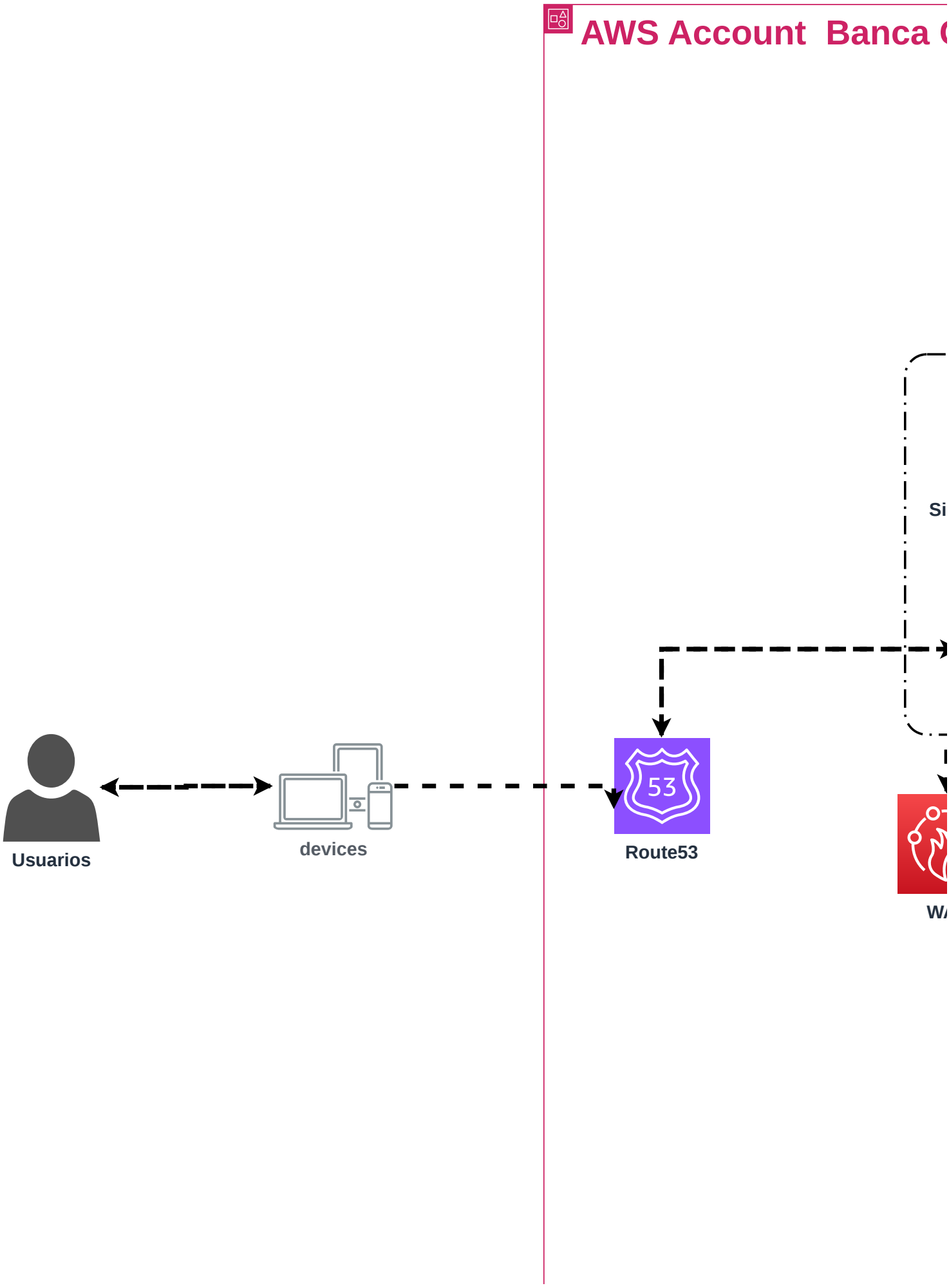

Netwrok ACL
72.16.0.0
72.16.1.0
72.16.2.0
T Database
Data subnet_2 (Private)
Security Group
Netwrok ACL
72.16.0.0
72.16.1.0
72.16.2.0
T Service
Service subnet_2 (Private)
Security Group
Netwrok ACL


Servers
Se
Direct Connect
TRANSIT GATEWAY
Router
Servers
S
VPN SITE-TO-SITE

ervers

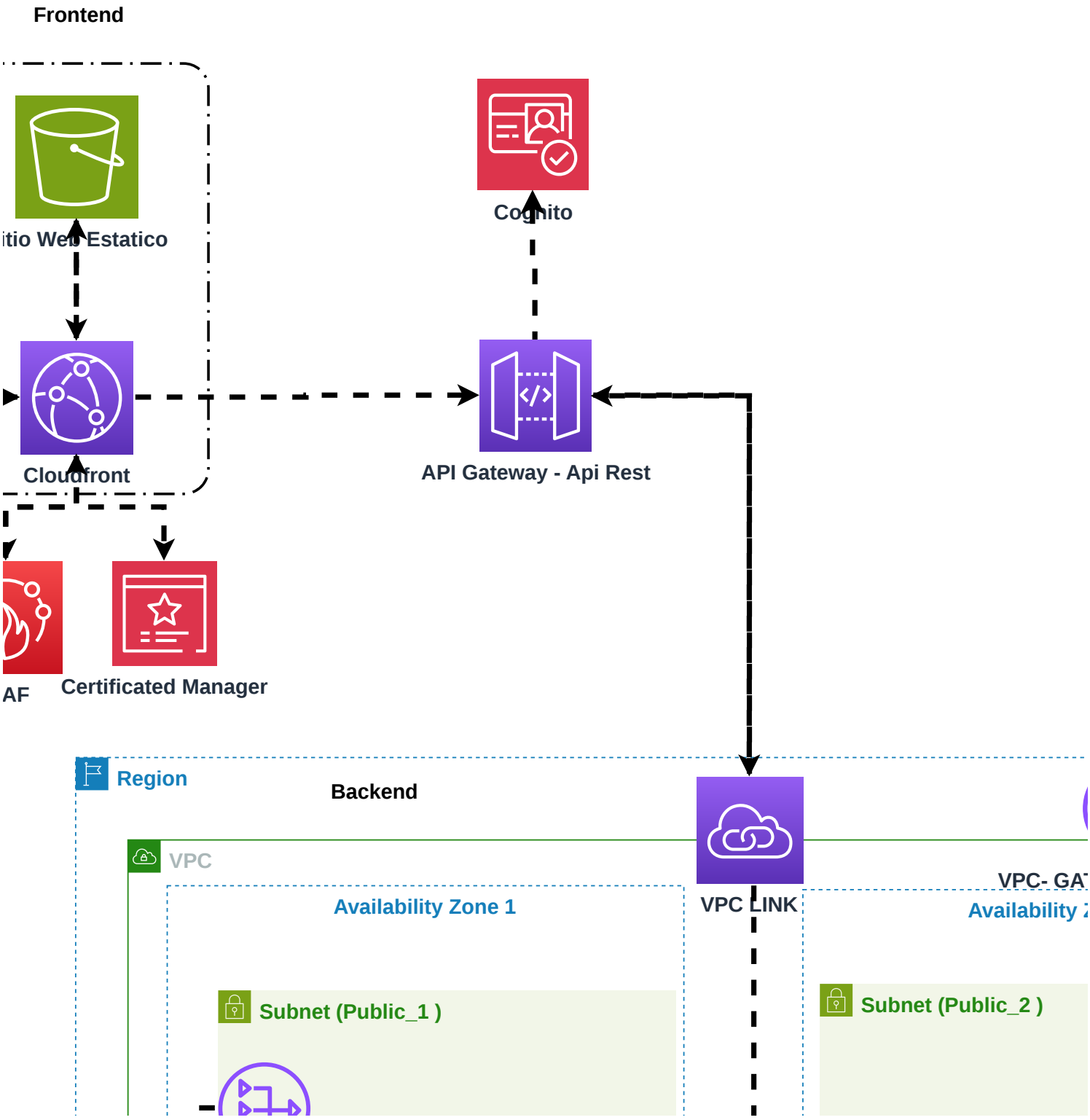
Servers

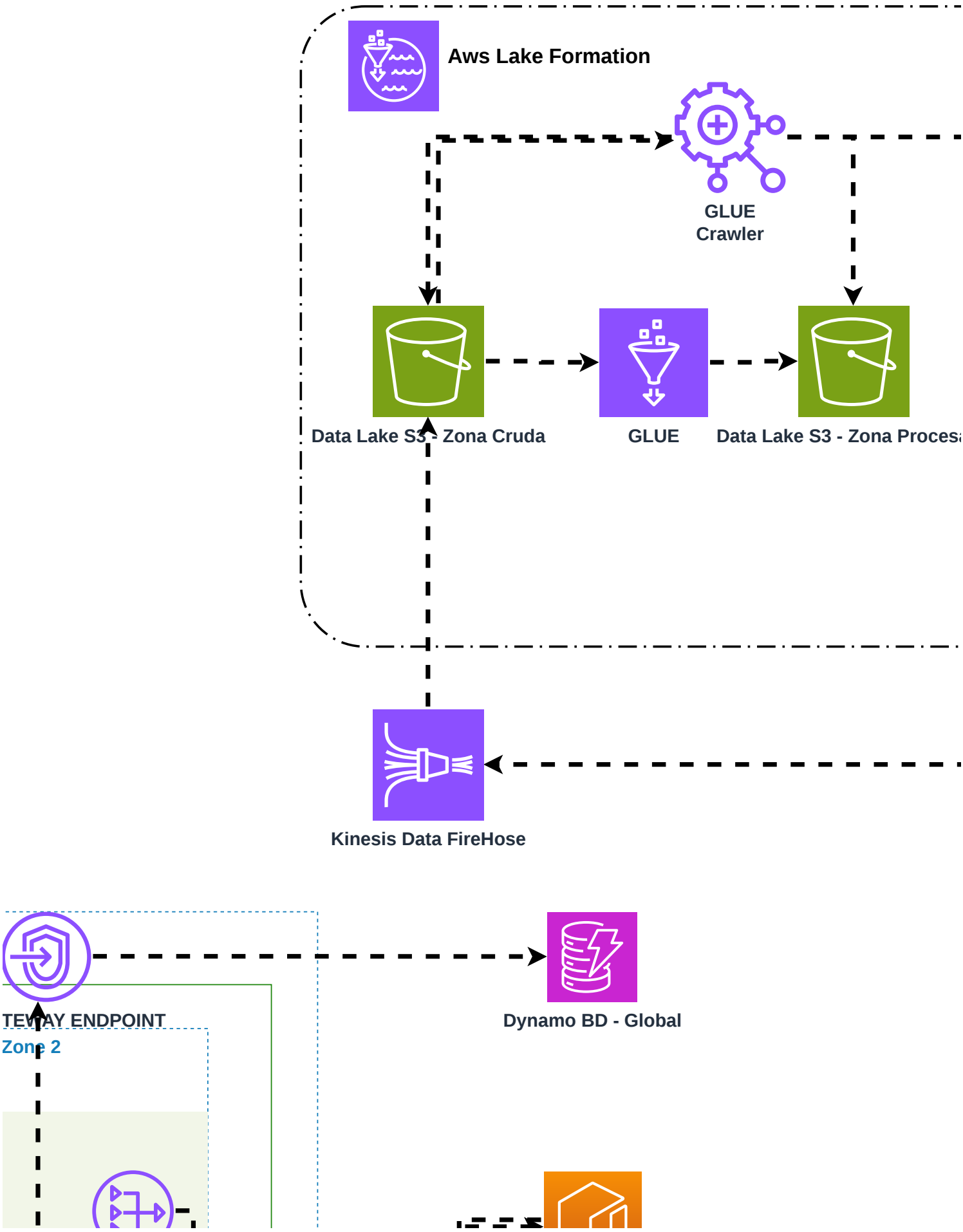
Infraestructura en AV

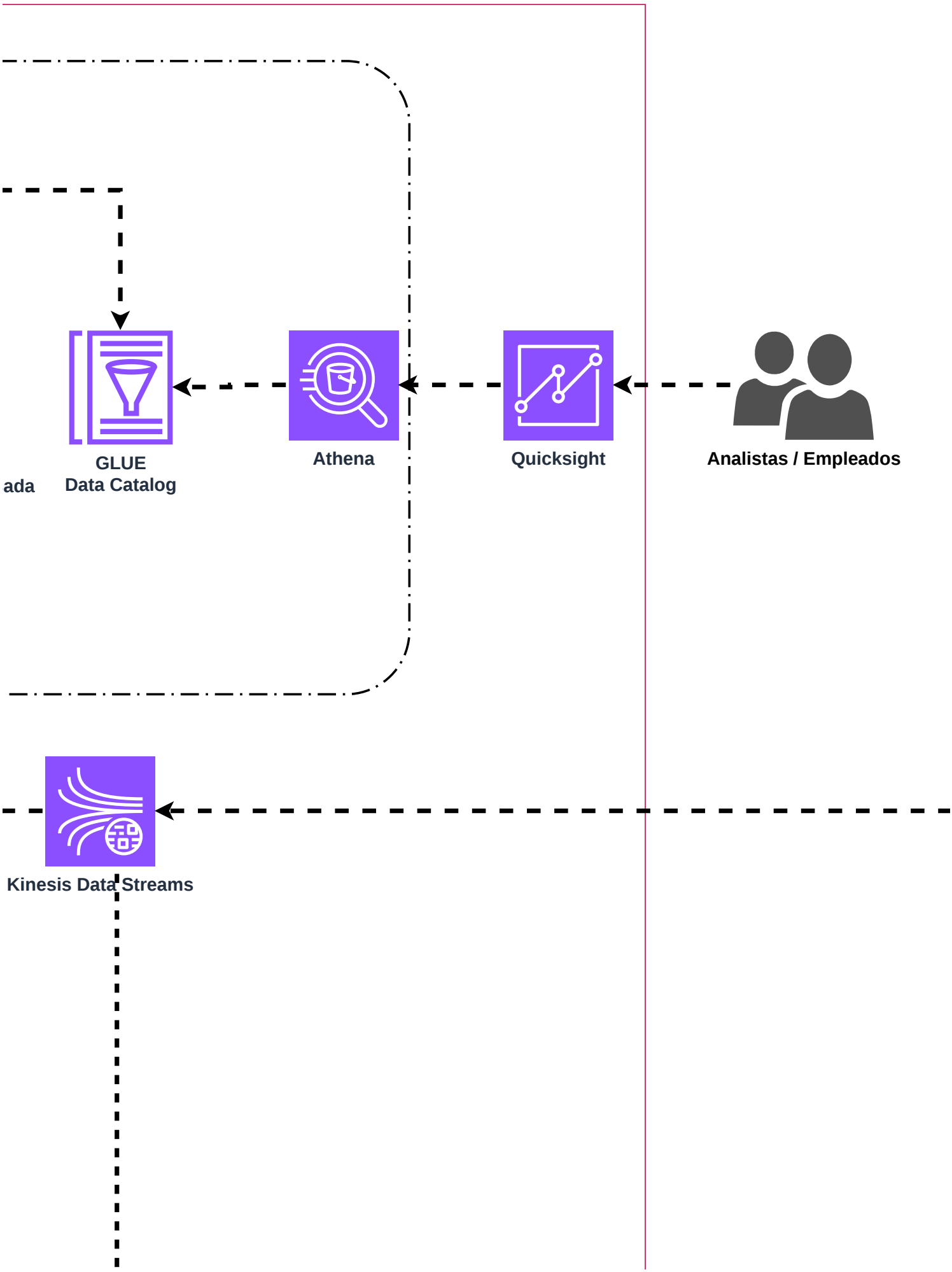


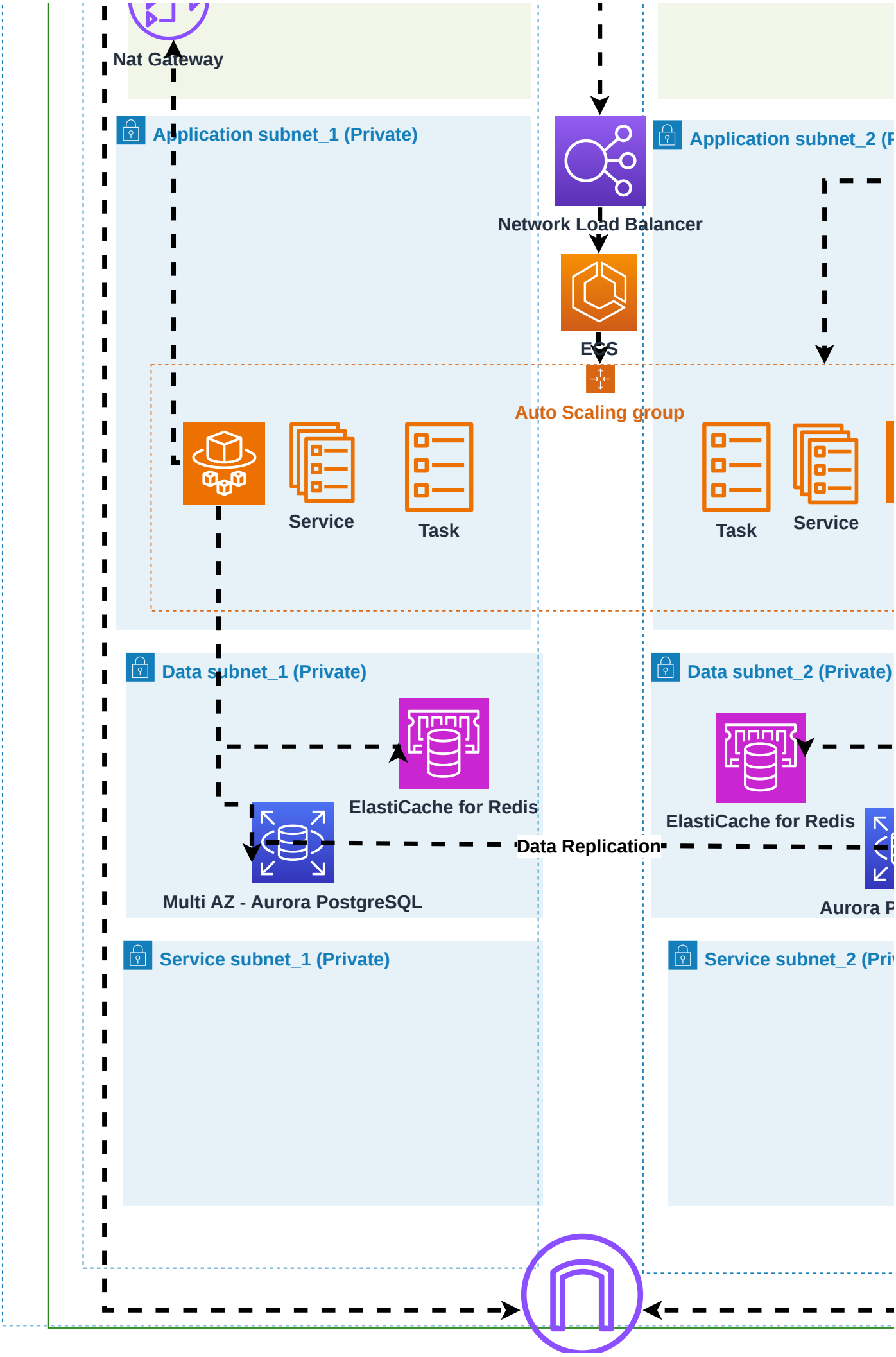
VS para Banca Global S.A.S

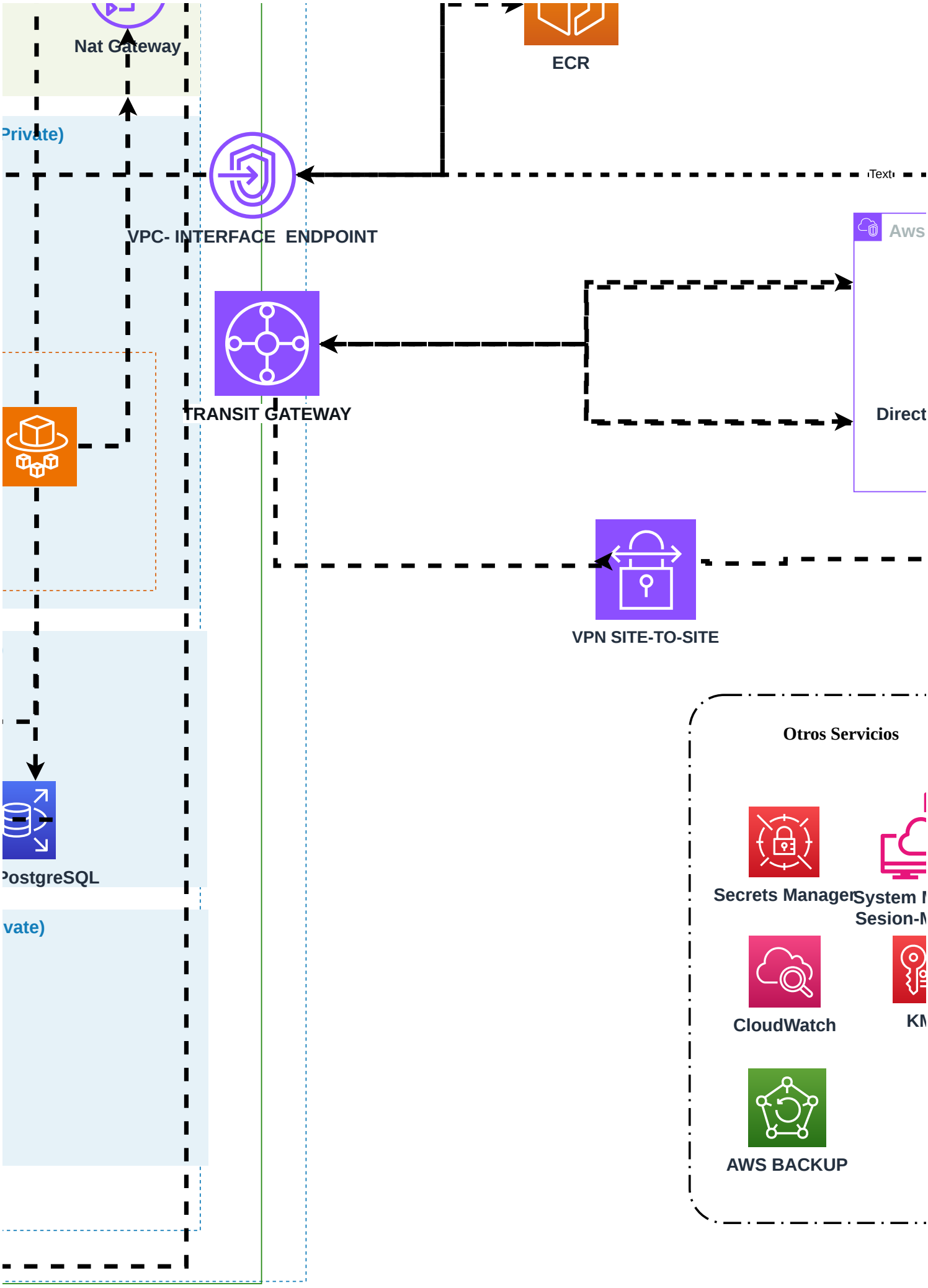
Global S.A.S

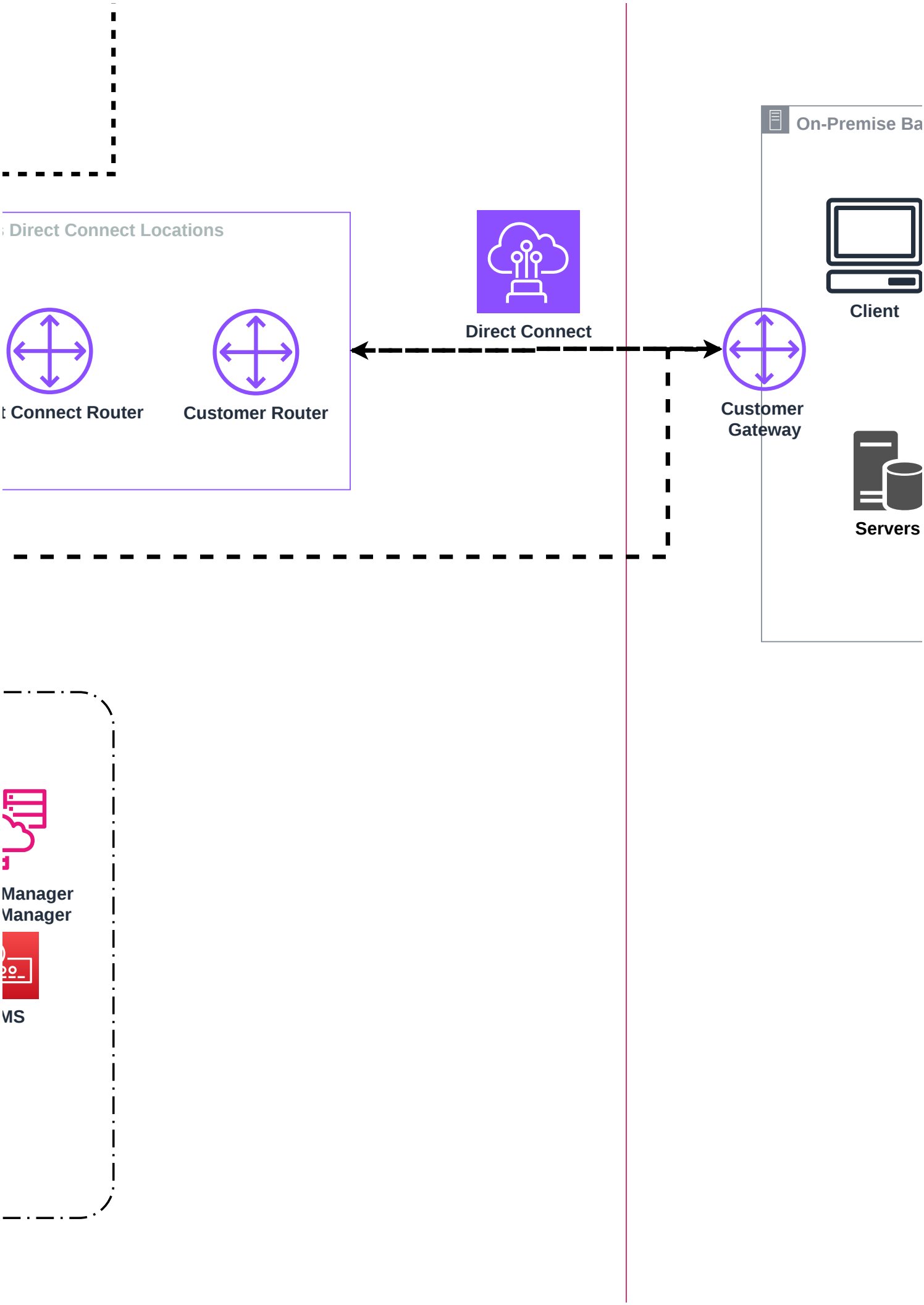


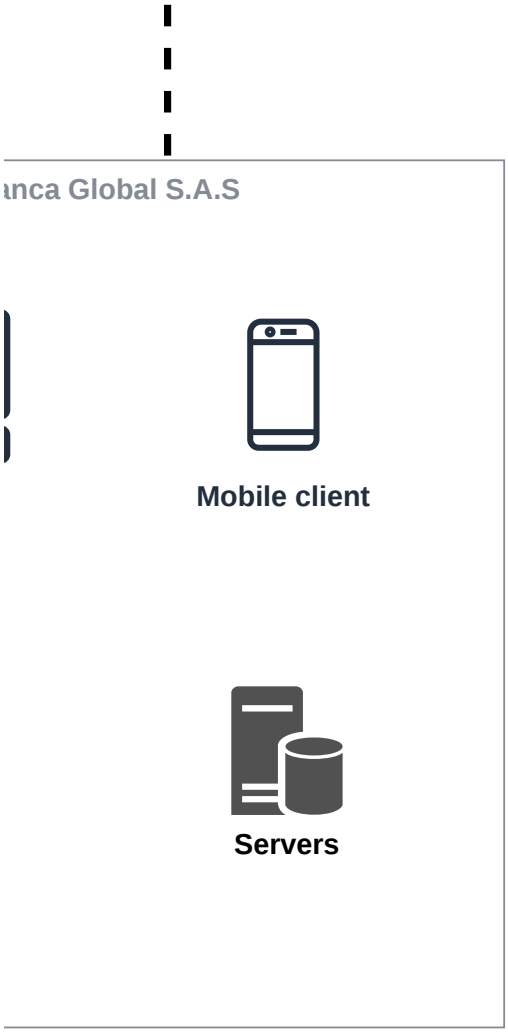

Aws Lake Formation
GLUE
Crawler
Data Lake S3 - Zona Cruda
GLUE
Data Lake S3 - Zona Proces
Kinesis Data FireHose
TEWAY ENDPOINT
Zone 2
Dynamo BD - Global

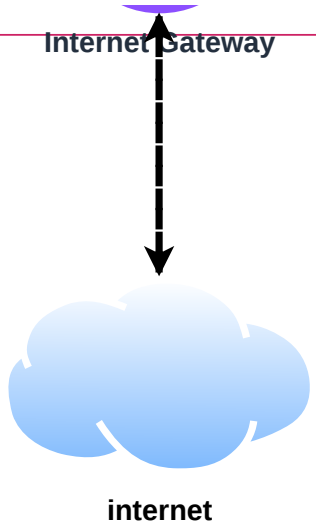

ada
GLUE
Data Catalog
Athena
Quicksight
Analistas / Empleados
Kinesis Data Streams

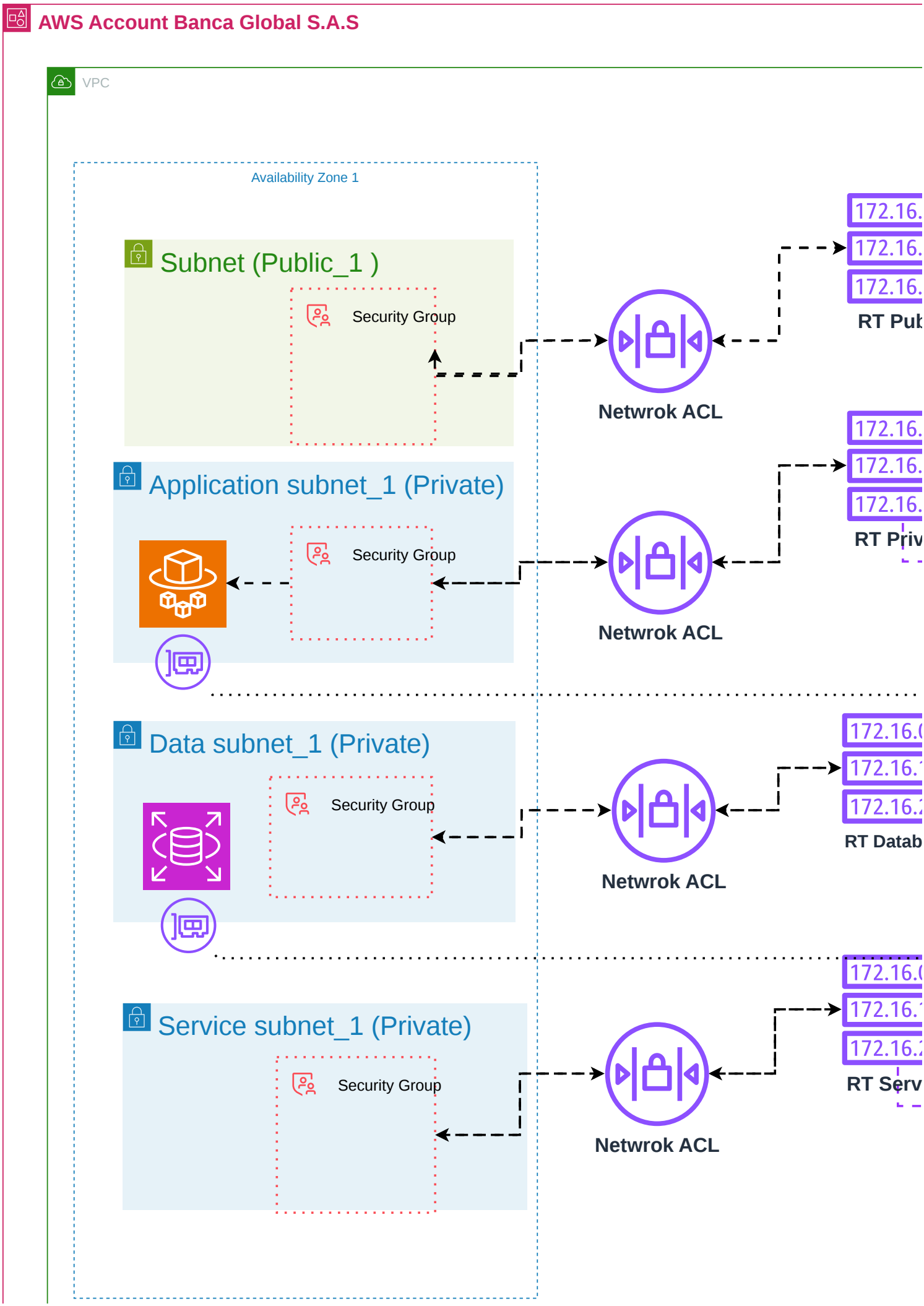

Nat Gateway
Application subnet_1 (Private)
Network Load Balancer
Application subnet_2 (P
ECS
Auto Scaling group
Service
Task
Task
Service
Data subnet_1 (Private)
Data subnet_2 (Private)
ElastiCache for Redis
ElastiCache for Redis
Data Replication
Multi AZ - Aurora PostgreSQL
Aurora P
Service subnet_1 (Private)
Service subnet_2 (Pri

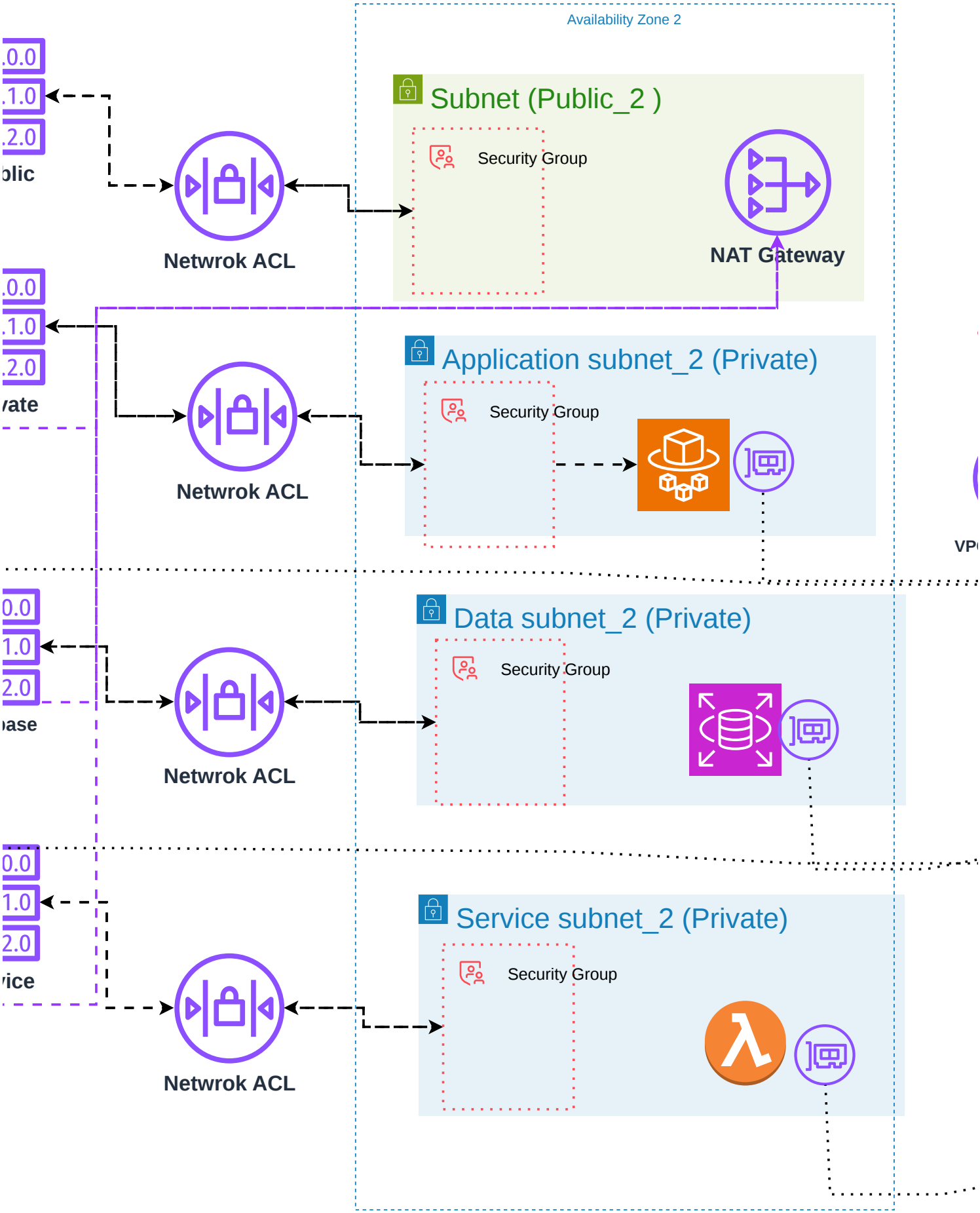

Nat Gateway
ECR
Private)
VPC- INTERFACE ENDPOINT
Aws
TRANSIT GATEWAY
Direct
VPN SITE-TO-SITE
PostgreSQL
vate)
Otros Servicios
Secrets Manager
System
Sesion-
CloudWatch
KM
AWS BACKUP

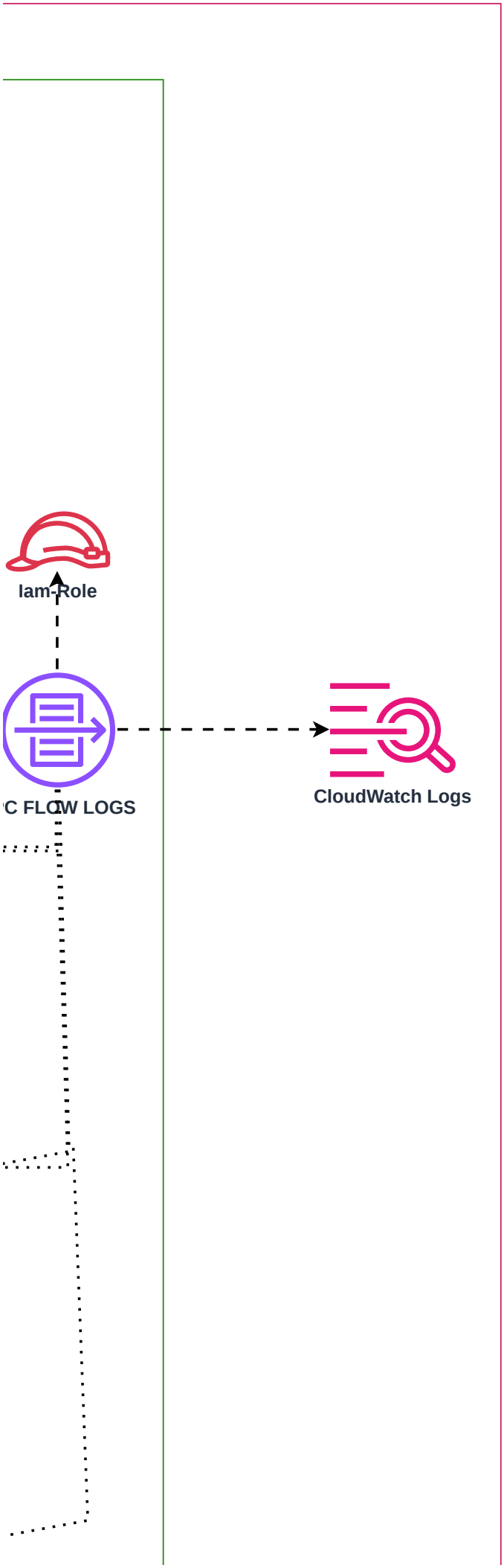

Direct Connect Locations
t Connect Router
Customer Router
Direct Connect
On-Premise Ba
Client
Customer Gateway
Servers
Manager
Manager
MS


anca Global S.A.S
Mobile client
Servers

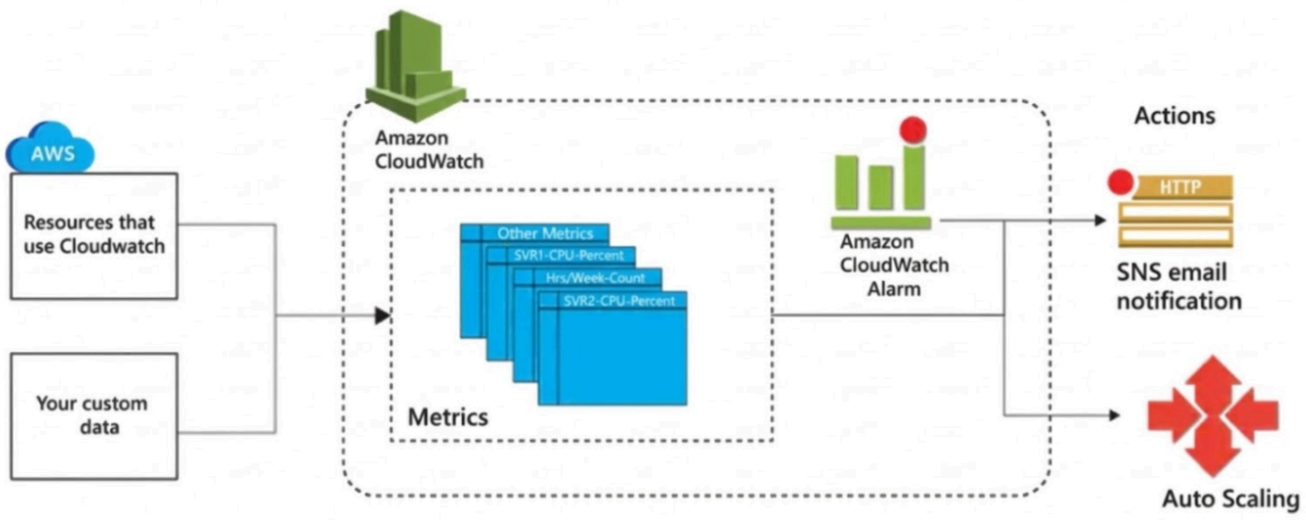

Internet Gateway
internet

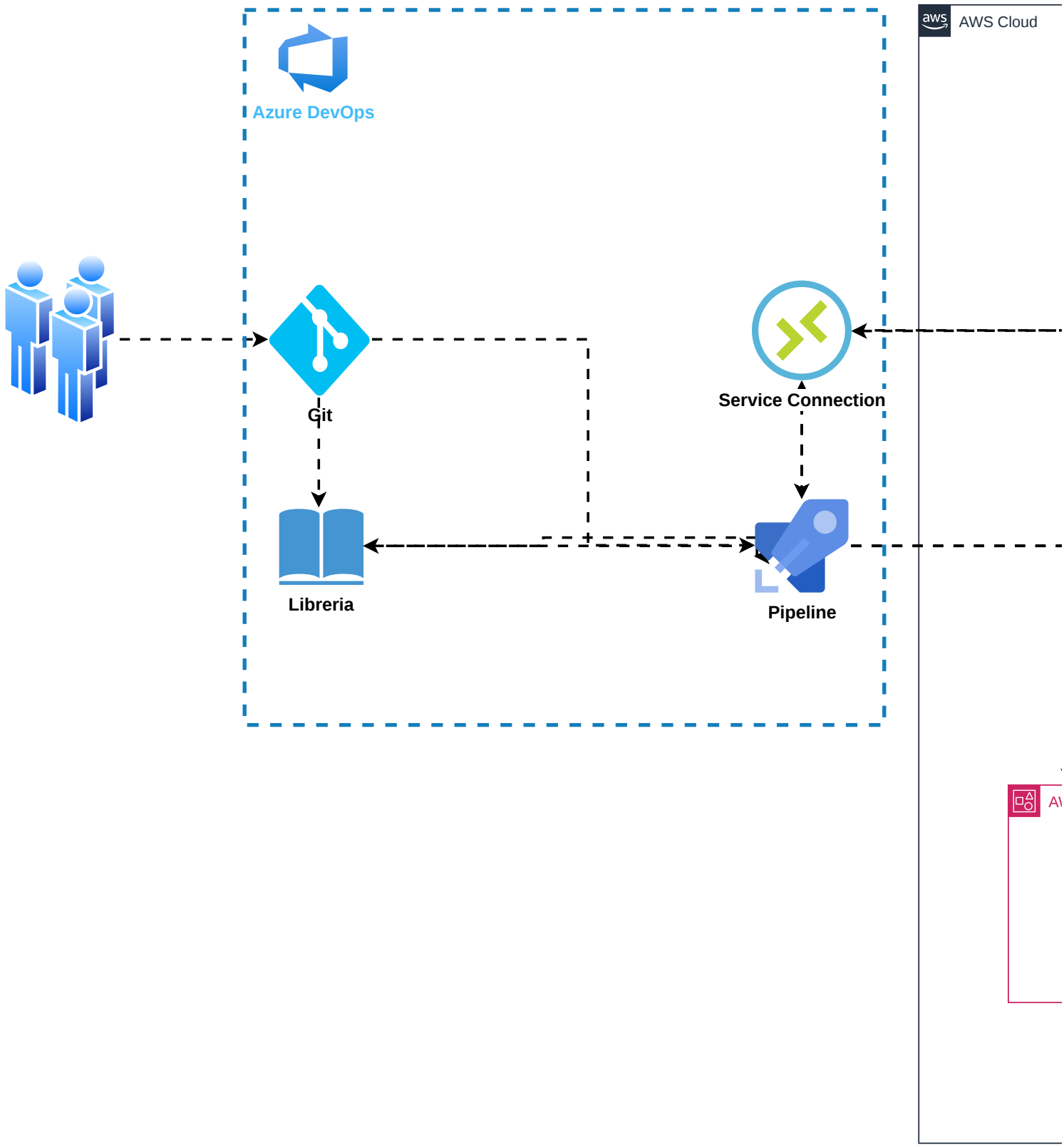

AWS Account Banca Global S.A.S
VPC
Availability Zone 1
Subnet (Public_1)
Security Group
Netwrok ACL
172.16.
172.16.
172.16.
RT Pub
Application subnet_1 (Private)
Security Group
Netwrok ACL
172.16.
172.16.
172.16.
RT Priv
Data subnet_1 (Private)
Security Group
Netwrok ACL
172.16.
172.16.
172.16.
RT Datab
Service subnet_1 (Private)
Security Group
Netwrok ACL
172.16.
172.16.
172.16.
RT Serv

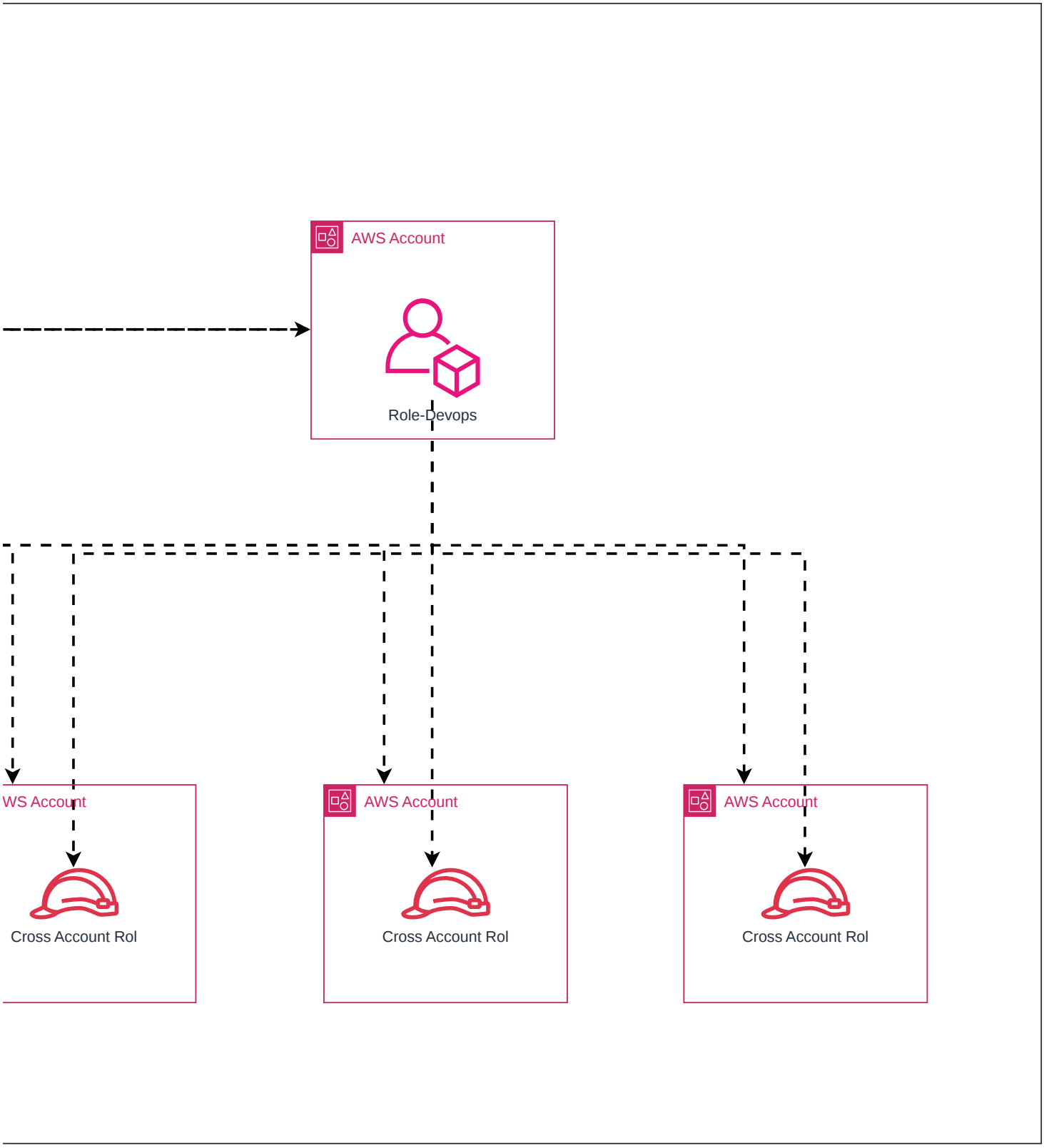

Availability Zone 2
Subnet (Public_2)
Security Group
NAT Gateway
Netwrok ACL
Application subnet_2 (Private)
Security Group
Netwrok ACL
Data subnet_2 (Private)
Security Group
Netwrok ACL
Service subnet_2 (Private)
Security Group
Netwrok ACL

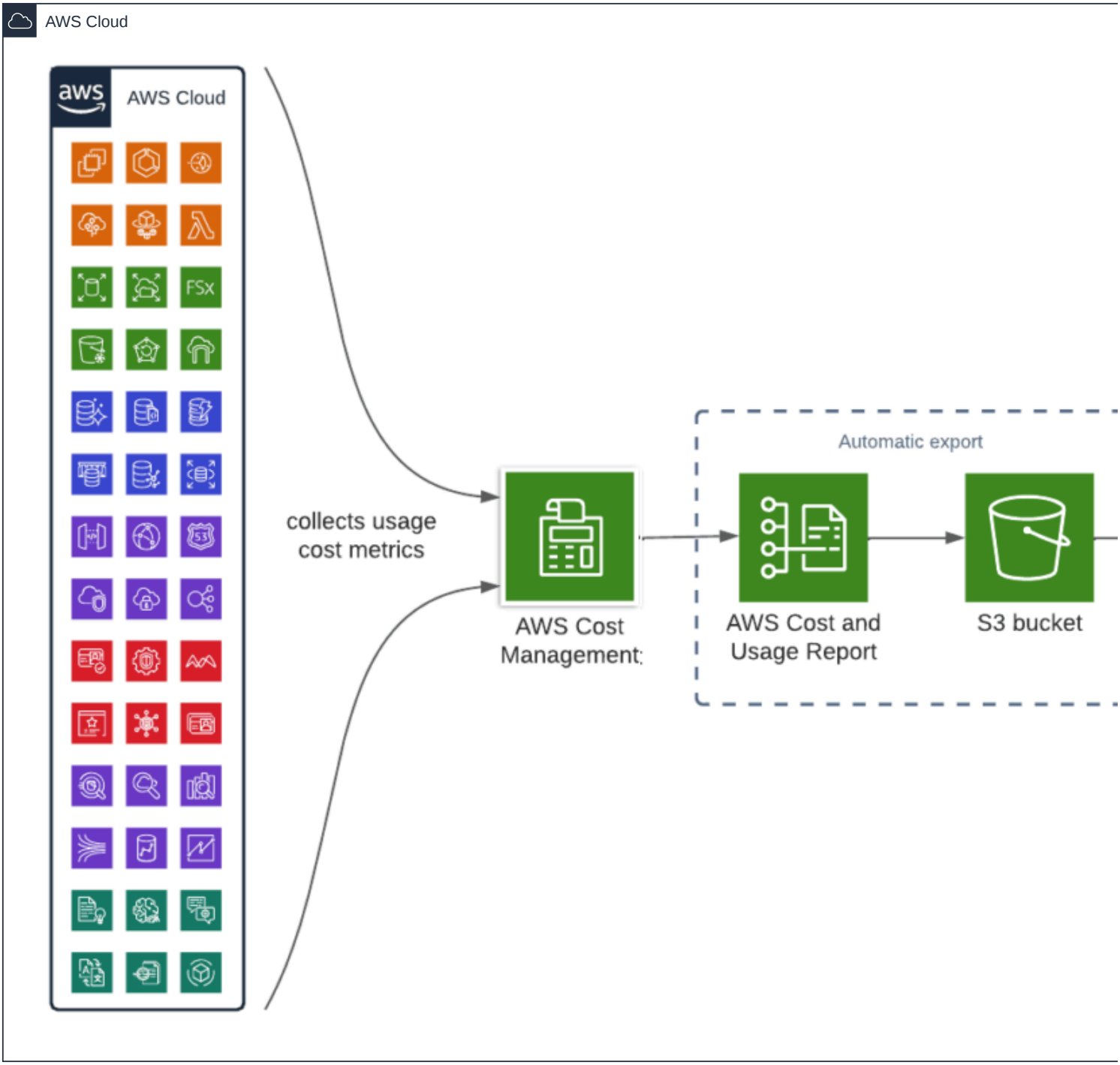

Iam-Role
C FLOW LOGS
CloudWatch Logs

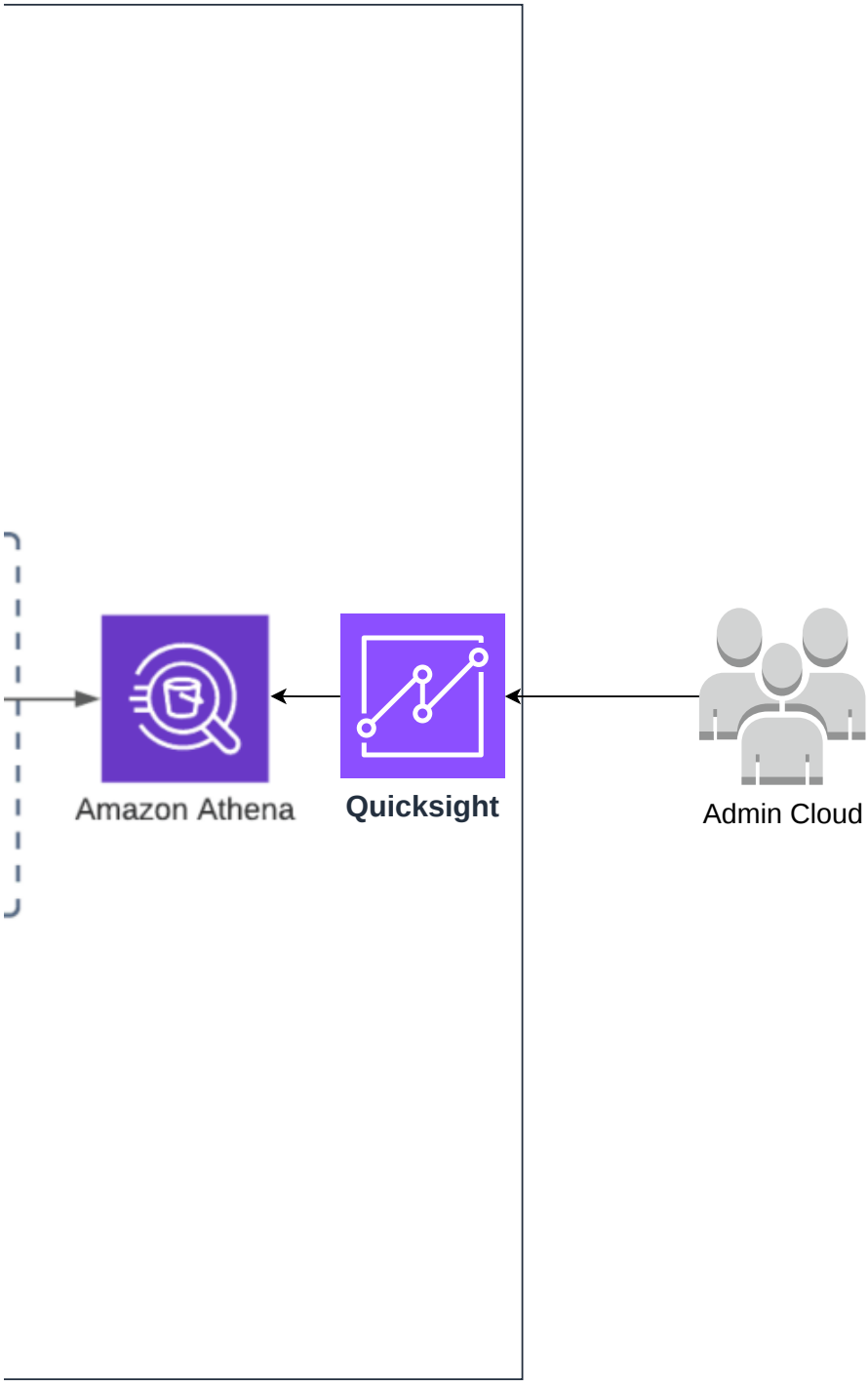
AWS Account Banca Global S.A.S

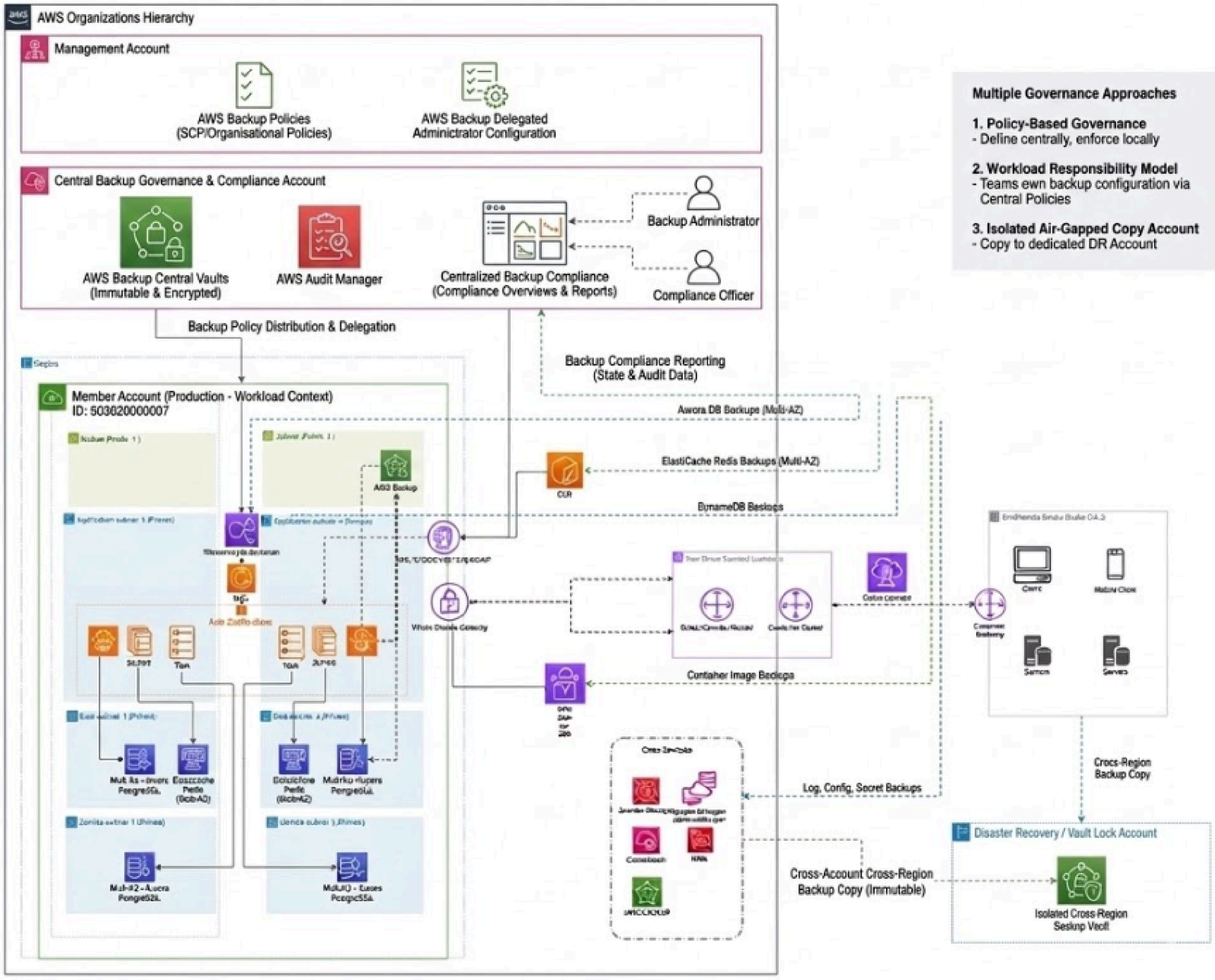



AWS Cloud
Azure DevOps
Git
Service Connection
Libreria
Pipeline


AWS Account
Role-Devops
WS Account
Cross Account Rol
AWS Account
Cross Account Rol
AWS Account
Cross Account Rol


AWS Cloud
AWS Cloud
collects usage
cost metrics
AWS Cost
Management
Automatic export
AWS Cost and
Usage Report
S3 bucket


Amazon Athena
Quicksight
Admin Cloud



Multiple Governance Approaches

1. Policy-Based Governance
- Define centrally, enforce locally

2. Workload Responsibility Model
- Teams ewn backup configuration via Central Policies

3. Isolated Air-Gapped Copy Account
- Copy to dedicated DR Account

Your Environment
Logging Without Limits
Live Search
Generate Metrics
Summarize and Track Trends
15-month Retention
Alert on Changes
Cloud SIEI
Out of the Box Rules for Threat Detection
STREAM ANALYCIS
Cloud Services
INGEST ALL LOGS
DATADOG
PROCESS, ENRICH & ROUTE
DISCOVER & REDACT SENSITIVE DATA
INTERACTIVE EXPLORATION
Automated Insights
Log Anomaly Detection
Error Tracking
Search and Analytics
Flexible Short & Long Term Retention
Metric-Trace-Log Correlation
Log Alerting
Unified Dashboards
Identify Advanc Log Patterns
LOG FORWARDING
Archives
Amazon S3
RENYDRATE HISTORICAL LOC
Custom Destinations
HTTP
Compliance Support
Audit Trail
RBAC | Supports HIPAA, FedRamp, PCI, and GovCloud
Available in multiple regions

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